1/11/2026

Software Requirement Specification

CampusCare

Version	Date	Authors
1.0	31 Aug 2026	Group F
2.0	13 Oct 2026	Group F
3.0	1 Nov 2026	Group F

WORK INTEGRATED LEARNING IN INFO TECHNOLOGY
Central University of Technology

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Group members

Group F	
Name and Surname	Student Number
BF Motseki	223038085
Lesedi D Mokheti	223051025
ML Mwenda	221034577
A Jara	223007530
TV Thabisi	223005893
S Bila	223048044
B Mbinga	223020021

Stakeholder and organisation

Organisation name and type

CUT Wellness Centre , University Wellness and Student Support Centre

Name	Role	Contact details
C.Maluka	Stakeholder	cmaluka@cut.ac.za
Dr N.Afrika	Stakeholder	nafrika@cut.ac.za

Contact person

Cleo Maluka – Senior Unit Administrator, Wellness Centre

Organisation's main activities

The Wellness Centre provides psychological counselling, health advice, medication related services within its scope of practice, and referrals to hospitals or specialists when required.

Reason for selecting the stakeholder.

The stakeholder was selected because they have professional knowledge of the Wellness Centre's services and processes and can provide valuable information about the challenges students experience when accessing wellness services. Their input helps the group understand the current appointment-booking process and identify requirements for improving the way students book appointments and how practitioners manage their availability and appointments.

Problem Statement

Many students face significant mental health challenges, stress and academic pressure. Cut wellness centre prioritises students' health and psychological needs they provide counselling services. According to the Wellness centre, student mental health concerns have increased substantially in recent years, yet many students who need support do not access it

Many students go to the wellness centre to book appointments to seek help. The organisation currently uses paper diaries and manual communication to complete student mental health service management. The administration has no digital tools to keep track student's progress. Students seeking help face stigma barriers, making it difficult to visit in person. Students have **no digital platform** for self-monitoring ,self-help, appointment reminders. The communication between the students and service providers Is limited to in-person. This results in duplicated bookings and scheduling errors, high no show rates. Inefficient use of therapist and admin time. Delayed support for students in crisis. No early warning system for at-risk students

1. **Purpose**

CampusCare aims to digitally transform mental health support services that the Central University of Technology by replacing manual, paper-based processes with an integrated mobile and web platform. Purpose is to support students through their academic journey. To provide students with **immediate, accessible, and private mental health support** through self-monitoring tools, peer connection, and emergency response, while empowering wellness staff with the visibility needed to **identify and support at-risk students early and offer post consultation support**—all within a secure digital platform.

2. **Definitions and abbreviations**

Term or abbreviation	Meaning
SRS	Software Requirements Specification
Flutter	Framework used to develop the mobile application
ASP.NET	Framework used to develop the web application
Supabase	Shared online backend and database

Project scope

Project Title:

CampusCare

CUT Student wellness appointment system
A mobile and web-based platform designed to provide university students with accessible mental health and wellness support.

Aim:

To design, develop, and deploy a secure, POPIA-compliant mobile and web platform that provides university students with accessible mental health support tools and connects them with CUT professional help when needed.

Objective:

- Enable students to track their mood and emotional wellbeing.
- Provide stress-management and mindfulness resources.
- Provide journaling and emotional check-in features.
- Encourage engagement through points, streaks and rewards.
- Enable secure communication and appointment scheduling with therapists/psychologists.
- Provide emergency support features.
- Protect student information through POPIA-compliant data processing.

Deliverable:

- Functional mobile application for iOS and Android.
- Functional web application with administrative features.
- UI/UX designs and prototypes.
- Database schema and ERD.
- API documentation.
- Integration testing and QA report.
- Security audit and POPIA compliance report.
- Technical documentation and user manuals.
- Final project presentation and demonstration.
-

Acceptance Criterion:

1. Both the mobile and web applications are successfully deployed.

2. The core wellness features are functional.
3. Secure therapist/student information
4. POPIA compliance has been verified.
5. Positive feedback is received from a pilot group of university students.
6. The final project is successfully demonstrated and defended.
7. Technical and user documentation has been completed.

Exclusion:

The system will not integrate with existing university systems.

Constraint:

- The project must be completed within 9 weeks.
- The budget is limited to free/open-source tools and services.
- There will be no integration with existing university systems.
- The mobile application must support both iOS and Android.

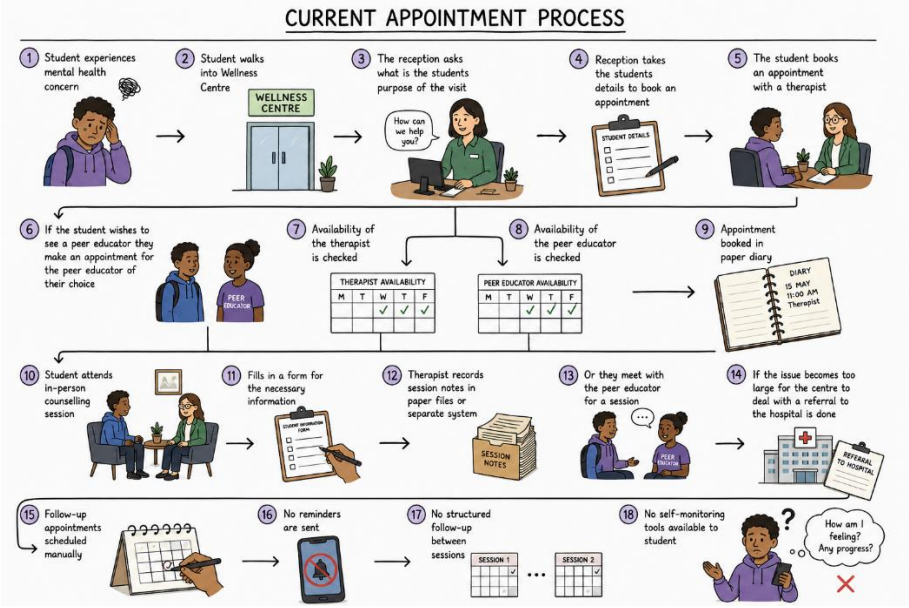
Assumption:

- Students will use their university email address for verification.
- Therapists/psychologists will be pre-registered by the university.
- Students will have access to a smartphone or computer with internet connectivity.

stake holder and user groups

Stakeholder or user	Role in the proposed system
External Stakeholder	Cleo Maluka
Mobile application user	Students
Web applications user	Therapist, peer educators, receptionist

Current Process



1. Student experiences mental health concern
2. walks into Wellness Centre
3. The reception asks what is the student's purpose of the visit
4. Reception takes the students details to book an appointment
5. The student books an appointment with a therapist
6. If the student wishes to see a peer educator, they make an appointment for the peer educator of their choice
7. Availability of the therapist and peer educator is checked
8. Appointment booked in paper diary
9. Student attends in-person counselling session
10. Fills in a form for the necessary information
11. Therapist records session notes in paper files or separate system
12. Or they meet with the peer educator for a session
13. If the issue becomes too large for the centre to deal with a referral to the hospital is done
14. Follow-up appointments scheduled manually
15. No reminders are sent
16. No structured follow-up between sessions
17. No self-monitoring tools available to student

Product perspective

The Student Wellness System is an **integrated digital platform** consisting of three interconnected components:

- **Flutter Mobile Application** – Student-facing app for mood tracking, journaling, appointment booking, and wellness resources.
- **ASP.NET Web Application** – Staff-facing dashboard for managing appointments, viewing student data, and identifying at-risk students.
- **Supabase Backend** – Shared database, authentication, real-time synchronisation, and file storage.

All three components operate as **one unified system** through the Supabase backend. Information entered through either platform is stored in the same database and synchronised instantly. This ensures:

- An appointment booked on Flutter appears on ASP.NET
- A mood entry logged on Flutter is visible to the therapist on ASP.NET
- Practitioner verification status is consistent across both platforms
- Student wellness data remains identical across both applications

Product functions

User Role	Functions
Students	Register/login, complete digital intake form, capture personal and next-of-kin information, provide consent, book/view/cancel appointments, log moods with streak history, complete emotional check-ins, create journal entries, access wellness resources, use emergency SOS, and manage their profile, access to community resources
Therapist	Log in, view verification status, manage availability, view appointments, view authorised student information (moods, journals, check-ins), and manage therapist records
Administrators	Manage users, manage and verify Therapist, manage appointments, view system reports, and access analytics
Peer Educators	Log in, view verification status, view health broadcasts, refer students to professional health

Functional Requirement, Non-functional, Integration, Prioritisation

Requirement Table

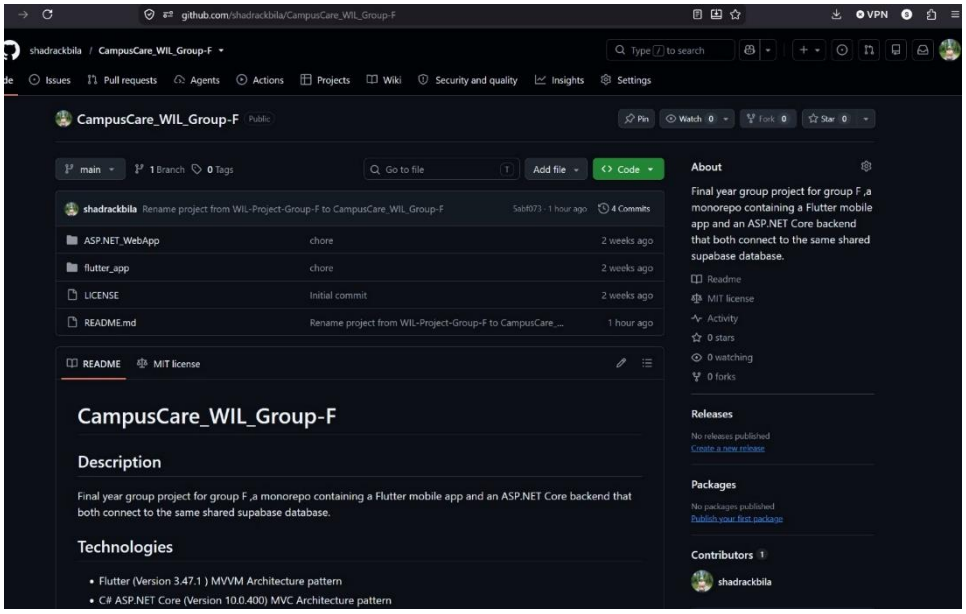
ID	REQUIREMENT	TYPE	PRIORITY
FR1	The student must be able to register and log in.	Functional	Must Have
FR2	The student must be able to complete the Wellness Centre Intake Form.	Functional	Must Have
FR3	The student must be able to provide next-of-kin information.	Functional	Must Have
FR4	The student must be able to select the purpose of the visit.	Functional	Must Have
FR5	The student must be able to provide consent.	Functional	Must Have
FR6	The student must be able to submit an appointment request.	Functional	Must Have
FR7	The student must be able to view and manage their appointments.	Functional	Must Have
FR9	The practitioner must be able to view authorized student information.	Functional	Must Have
FR10	The administrator must be able to verify practitioners.	Functional	Must Have
FR11	The student must be able to create, update, and delete their own mood entries.	Functional	Must Have
FR12	The student must be able to create, update, and delete their own journal entries.	Functional	Must Have
FR13	The student must be able to manage their own emotional check-ins.	Functional	Must Have
NFR1	Only authorized users may access protected student information.	Non-functional	Must Have
NFR2	Practitioners must only be able to view authorized student wellness records.	Non-functional	Must Have
NFR3	Practitioners must not be able to create, update, or delete student-owned wellness records.	Non-functional	Must Have
SR1	The mobile application must be developed using Flutter.	System	Must Have
SR2	The web application must be developed using ASP.NET/C#.	System	Must Have
IR1	Information submitted through Flutter must be stored in Supabase and available in ASP.NET where authorized.	Integration	Must Have
IR2	Information submitted through ASP.NET must be stored in Supabase and available in Flutter where authorized.	Integration	Must Have
IR3	Appointment updates made through either platform must be reflected on the other platform.	Integration	Must Have
IR4	Student wellness data permissions must be enforced consistently across both platforms.	Integration	Must Have

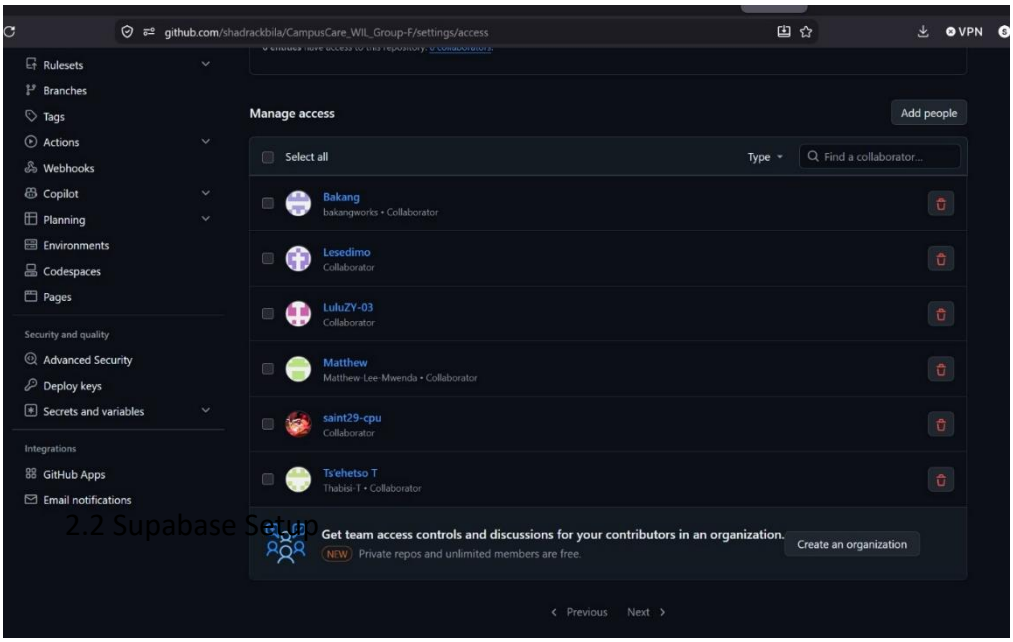
FR14	The student should be able to use medication reminders.	Functional	Should Have
FR15	The system should provide streaks and achievements.	Functional	Should Have
FR16	The system should provide additional reporting.	Functional	Should Have
IR1	Information submitted through Flutter must be stored in Supabase and available in ASP.NET where authorized.	Integration	Must Have
IR2	Information submitted through ASP.NET must be stored in Supabase and available in Flutter where authorized.	Integration	Must Have

Project Setup

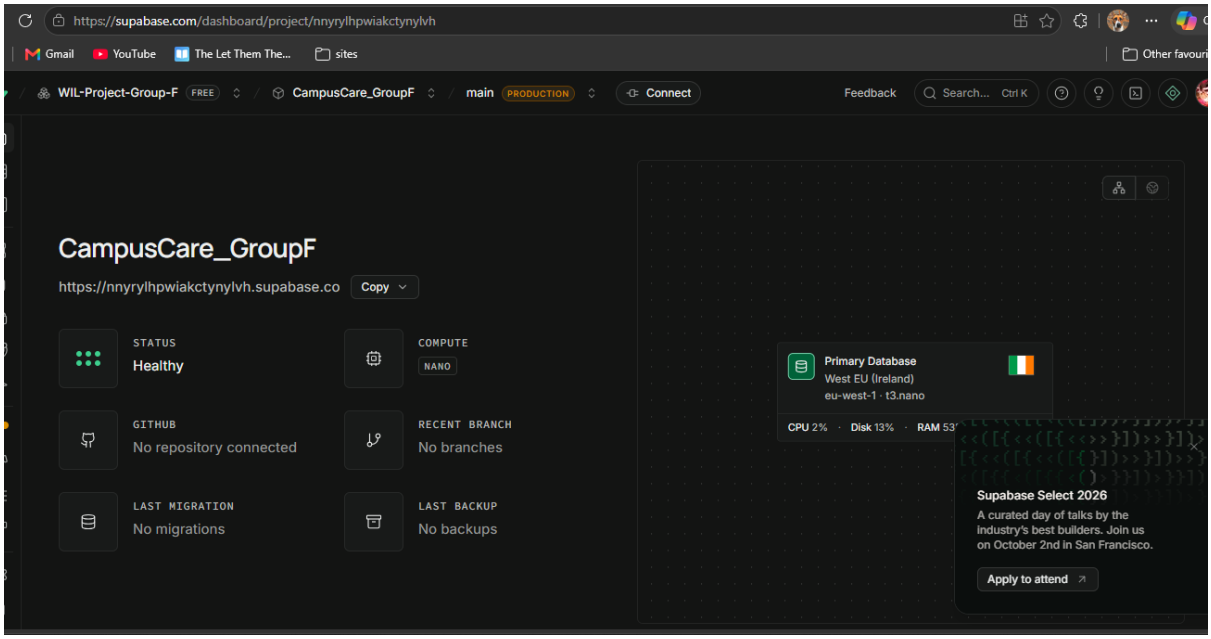
2.1 GIT HUB

Git Hub Link : https://github.com/shadrackbila/CampusCare_WIL_Group-F.git





2.2 Supabase Setup



Stakeholder Evidence

Stakeholder Interview			
Date:	20 August 2026	Time	12:05 - 12:25
Location	CUT Wellness Centre offices		
Role	Responsibility	Group Member	
Facilitator	Leads the interview	A Jara	
Note-taker	Records responses	TV Thabisi	
Timekeeper	Monitors the meeting	LD Mokheti	
Requirements Checker	Summarizes and confirms the needs	BF Motseki	
Observer	Observes stakeholder body language and non-verbal cues; notes unspoken concerns	S Bila	
Technical Advisor	Explains technical concepts in simple terms; answers technical questions about the system	ML Mwenda	
Stakeholder Liaison	follows up after the meeting	B Bakang	

Interview Questions and Stakeholder responses

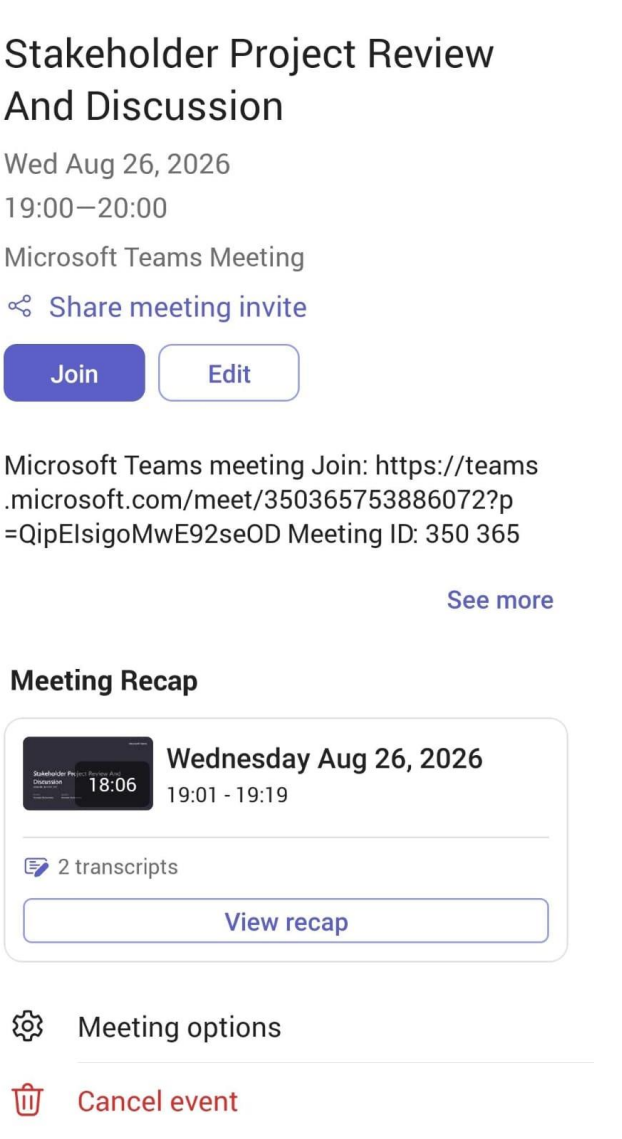
Question	Stakeholder Response	Follow-up Required	Confirmed?
How does the current appointment booking process work?	Students currently book appointments by physically visiting the Wellness Centre. The process is largely manual.	No	Yes
1. What services does the Wellness Centre provide?	Psychological counselling, Health advice. Medication-related services within the center's scope of practice. Referrals to hospitals or specialists when a student's condition falls outside the Centre's scope of practice	Yes	Yes
2. What problems or challenges are currently experienced?	The manual process is an inconvenience for students. As they must physically book an appointment which poses as an inconvenience on days, they are not on Campus and must travel to simply book an appointment. Another is that students do not have a centralized way to see which psychologists/counselors are available.	No	Yes
3. What information is required when making an appointment?	Students provide information such as their name, student number, and contact details. There are also confidentiality and consent requirements that need to be communicated to students before receiving certain services.	No	Yes
4. What services should the proposed system support?	The Wellness Centre provides both psychological counselling services and healthcare services. Further consultation with the clinic general manager will be required to understand their specific challenges and to determine whether the services should be incorporated into one system or treated as separate components.	Yes	Yes
5. What information should students see before booking?	Students should be able to see the available practitioners and appointment slots, including practitioner availability.	No	Yes

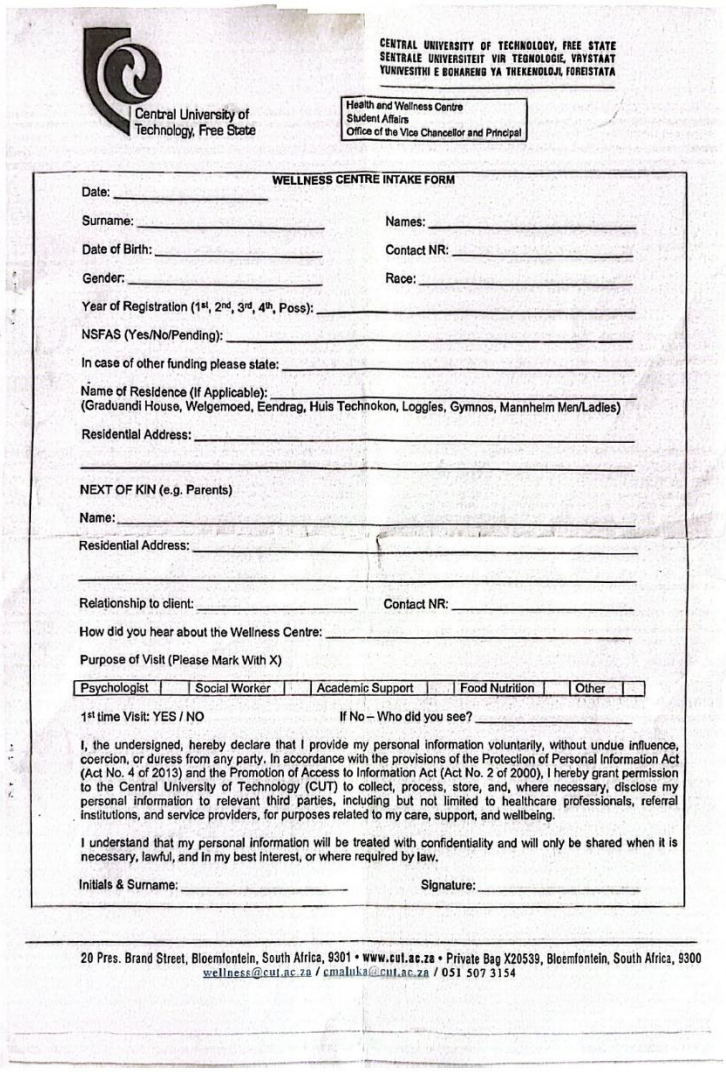
	And should be able to select the preferred gender of the practitioner.		
6. What security requirements are important?	Patient confidentiality and protection of sensitive information are critical. Real patient data should not be accessible to the development team	Yes – Consult with CUT IT Department	Yes
7. Do you have any specific expectations from the project?	The stakeholder explained CUT's goal is to move from paper-based processes towards becoming a paperless institution by 2030. Therefore, the project is expected to support this broader technological transformation. The key expectation is that the system should digitize the appointment booking process while improving convenience and accessibility for students.	No	Yes
8. What do you think would derail this project?	A breach of patient confidentiality or violation of institutional privacy policies could significantly affect or derail the project. Technical glitches, system failure or security problems could also negatively affect the project	Yes	Yes
9. Would you be willing to participate in future project reviews?	Yes. The stakeholder is willing to review the proposed solution and provide feedback, advice, corrections, and additional requirements.	No	Yes
10. How should we communicate with you for project reviews?	The stakeholder communicated that they can be contacted through: • Direct phone call • WhatsApp	No	Yes

Stakeholder Follow up meeting			
Date:	26 August 2026	Time	17:01 – 17:20
Location	Teams Meeting: Follow Up meeting		
Role	Responsibility	Group Member	
Facilitator	Leads the interview	A Jara	

Note-taker	Records responses	TV Thabisi
Timekeeper	Monitors the meeting	LD Mokheti
Requirements Checker	Summarizes and confirms the needs	BF Motseki
Observer	Observes stakeholder body language and non-verbal cues; notes unspoken concerns	S Bila
Technical Advisor	Explains technical concepts in simple terms; answers technical questions about the system	ML Mwenda
Stakeholder Liaison	follows up after the meeting	B Bakang

Meeting Evidence





Scanned with CamScanner

Feasibility Study

2.3 Technical Feasibility

Technical Requirements

The proposed Student Wellness System will consist of a Flutter mobile application and an ASP.NET/C# web application, with both platforms using Supabase as the shared backend/database. The system will allow information submitted through either platform to be stored and accessed by authorized users across both platforms. The technical feasibility assessment considered the group's available skills, development technologies, hardware, testing devices, internet connectivity and the complexity of the proposed functionality.

Technical feasibility Findings

Area	Finding
Flutter development	The group will use Flutter for the mobile application.
ASP.NET/C# development	The web application will be developed using ASP.NET/C#.
Supabase	Supabase will be used as the shared database/backend.
System integration	Flutter and ASP.NET will connect to the same Supabase backend
Development hardware	Development computers are required for both Flutter and ASP.NET development.
Testing devices	Smartphones and computers will be required to test the applications.
Internet connectivity	Internet access is required for development, Supabase and stakeholder communication.
Skills development	The group may need to strengthen skills in Supabase integration, authentication, role-based access and secure data handling.
Technical challenges	Secure communication, emergency support, role-based access and synchronisation between the two platforms may require additional development effort.

Technical Feasibility Decision

Decision: Feasible with changes

The system is technically feasible because the required technologies have been identified and the Flutter and ASP.NET applications can use the same Supabase backend. However, some functions may be technically challenging for the group and may require additional learning, testing and prioritization.

2.4 Operational Feasibility

The proposed system is intended to address the current manual appointment-booking process at the CUT Wellness Centre. Currently, students must physically visit the Wellness Centre to make appointments, and there is no centralized digital platform for viewing practitioner availability and appointment slots. In addition, practitioners do not have an efficient digital system to manage their clients, appointments, availability, and relevant student information, resulting in greater reliance on manual processes.

The proposed system will support students, practitioners and administrators, with different functionality based on their roles

Intended Users

User	Main Responsibilities/Functions
Students	Register, complete intake forms, provide consent, book and manage appointments, complete wellness activities and manage their profiles.
Practitioners	Manage their availability, view and manage appointments, access authorised student information, and manage relevant practitioner and session information.
Administrators	Manage users, practitioners, practitioner verification, appointments and reports.

Operational Feasibility Findings

Area	Finding
Problem addressed	The system will digitise the manual appointment-booking process.
User suitability	Students, practitioners and administrators are the intended users.
Device requirements	Students, practitioners and administrators are expected to access the system using smartphones or computers.
Internet requirements	Internet connectivity will be required to access the system.
Organisation fit	The solution supports CUT's goal of moving from paper-based processes towards a paperless institution.
Training	Staff and administrators may require basic system training.
Privacy	Access to sensitive student information must be restricted to authorised users.

The stakeholder specifically stated that the system should support CUT's move towards becoming a paperless institution by 2030.

14.3 Economic feasibility

The economic feasibility assessment focuses on whether the group has access to the resources required to develop and demonstrate the prototype.

Resource	Availability	Action Required
Internet/Data	Available	Team members need ensure they have sufficient personal data for development and meetings. Unless utilize the university free data.
Transport	Available	Arrange transport for stakeholder meetings when required.
Development computers	Available	Use available computers for development.
Test devices	Available	Use smartphones and computers for testing.
Development software	Available	Use required/free development tools.
Supabase	Available	Use Supabase for the shared backend/database.
Presentation resources	Available	Use available university/personal resources.
Other resources	Available	Use existing group resources where possible.

Economic Feasibility Decision

Decision: Feasible

The project is economically feasible for the proposed prototype because the group can use existing development resources and free/open-source tools and services. The project constraints specify that the budget is limited to free/open-source tools and services.

14.4 Schedule feasibility

Evaluate Schedule Feasibility

Question	Group finding
Is enough time available for system design?	**Yes.** The original time of 3 weeks allocation in the beginning was enough, but we fear that the time is shorten time will be tight. Since it reduced to 2 weeks

Is enough time available for Flutter development?	Yes ,with focus, with the allocated roles and everybody doing and completing their tasks it will be developed in time
Is enough time available for ASP.NET development?	**Yes, with focus.** We have 3 weeks of development
Is enough time available for Supabase integration?	Yes. Supabase setup, authentication, database design, and real time communication can be completed within 2-3 weeks, integrated alongside development
Is enough time available for testing and corrections?	**Yes.** Last week of development, allocated for manual testing, stakeholder feedback, and corrections. 1 week is tight but manageable for MVP.
Is enough time available for stakeholder feedback?	**Yes.** Communication with the stakeholder is simple and quick, a meeting will be scheduled ahead of time
Are too many functions currently included?	No, we actually want to add more
Which functions may need to be simplified, postponed or removed?	No, the feature makes the project

Schedule Feasibility Decision

The project is achievable within the 8 weeks semester **provided we prioritise core features**. Development time has been reduced (system design now 2 weeks instead of 3), which is tight but manageable with clear role allocation. Adding more features increases risk of incomplete delivery. We will deliver MVP features first and treat additional features as stretch goals.

14.5 Legal feasibility

Evaluate legal Feasibility

Question	Group finding
Will the prototype collect personal information?	Yes. Student name, email, student number, mood logs, journal entries, appointment history, messages
Is all the information necessary?	Yes. All data collected serves a functional purpose: identification (email/student number), mental health tracking (mood/journal), appointment management, and communication (messages)
Who should be allowed to access the information?	Students, Therapist, Admin for data verification and management
Will test data be used during development?	**Yes.** All development and testing will use anonymised data. No real student data will be used.

How will passwords and private keys be protected?	Passwords hashed using Supabase Auth
Does the project involve confidential stakeholder information?	No is does not. The Wellness Centre is providing general process information only.
Is private stakeholder information being kept out of GitHub?	Yes

Legal Feasibility Decision

The project is legally feasible. All data collected is necessary and consensual. Only test data will be used during development. Passwords are hashed using Supabase Authentication. No confidential stakeholder information is involved. Sensitive credentials will be excluded from GitHub. POPIA compliance will be maintained (consent, access rights, data minimisation).

14.5 Ethical Feasibility

Ethical Feasibility

Question	Group finding
Does the stakeholder understand the purpose of the project?	**Yes.** Stakeholder has been clearly informed it is an **educational prototype**. They agreed to participate.
Is unnecessary personal information being collected?	**No.** Only necessary information: email (verification), student number (identity), mood/journal/messages (core functionality). No sensitive health data without consent.
Could the proposed solution negatively affect intended users?	No, the risk is low, it provides
Is information being used only for its intended purpose?	**Yes.** Informed consent at registration. Opt-in/out for data sharing. Right to access, correct, and delete data (POPIA). The data is anonymous for analysis.

Summary of evaluation

The project is ethically feasible. Stakeholder understands it is an educational prototype and has agreed to participate. Only necessary data is collected. Informed consent is obtained at registration with opt-in/out options for data sharing. Users have rights to access, correct, and delete their data. The risk of negative impact is low as the app provides supportive tools.

Feasibility conclusion

Feasibility Area	Decision	Main Reason	Change Required
Technical	Feasible with changes	Some advanced features require additional technical skills and development time	conduct early proof-of-concept
Operational	Feasible	The system directly addresses Wellness Centre's current problems and fits the intended environment	Schedule regular stakeholder feedback meetings
Economic	Feasible	The prototype can be developed using available resources: Supabase free tier, Flutter, ASP.NET open-source tools. No major costs involved.	None
Schedule	Feasible with changes	Timeline is tight (design reduced to 2 weeks), but manageable with clear role allocation and scope prioritisation	Prioritise MVP features; treat additional features as stretch goals
Legal	Feasible	POPIA compliance planned. Only fictional/test data used. No sensitive info on GitHub. Passwords hashed.	Ensure consent forms and data deletion functionality are included
Ethical	Feasible	Stakeholder informed. Data minimisation. Informed consent with opt-in/out. Low risk to users.	Test emergency features thoroughly

Overall Feasibility Decision

The proposed Student Wellness System is **feasible with some changes**. It addresses the Wellness Centre's manual processes while adding wellness features like mood tracking, journaling, messaging, and student verification.

Core MVP features (appointment booking, mood tracking, journaling, secure messaging, student verification, emergency panic button, and admin dashboard) will be delivered within the 9-week semester. Additional features will be developed only if time permits.

We will prioritise Must Have requirements and postpone lower-priority features where necessary. The project is technically, operationally, economically, legally, and ethically sound. All development and testing will use fictional/anonymised data only.

Decision: Feasible with changes

Score Scale:	Level
>=3	Low
>=6	Medium
>=9	High

Project Risk Register

Risk ID	Risk Statement	Category	Likelihood (1-3)	Impact (1-3)	Score	Level	Mitigation Notes	Contingency	Risk Owner	Status
R01	The project team may not fully document the stakeholder's current Wellness Centre process and needs, which could result in incomplete or inaccurate requirements.	Requirements	2	3	6	Medium	Compare the stakeholder interview findings with the current Wellness Centre process and review the findings as a group before finalizing requirements.	Re-engage with the stakeholder to clarify missing information and update the requirements before the final approval.	LD Mokheti	Open
R02	The project team may incorrectly define the system scope. Resulting in features being included that are not required or feasible within the project timeframe.	Scope	3	3	9	High	Review the proposed features against the stakeholder needs, feasibility findings and the available project timeframe before approving the scope.	Remove or postpone unnecessary features and revise the scope to focus on the required and feasible functionality.	BF Motseki	Open
R03	Development tasks may take longer than planned, leaving insufficient time for testing, corrections and final demonstrations.	Requirements and Documentation	2	3	6	Medium	Review the requirements against the stakeholder findings, the current process and project scope before approving the SRS.	Conduct additional requirements review and update the SRS before moving into the detailed design.	LD Mokheti	Open
R04	The feasibility analysis may overlook important technical, resource or time constraints, which could result in requirements being selected that the team cannot realistically implement.	Feasibility	2	3	6	Medium	Evaluate the technical, resource and time constraints before finalising the requirements and proposed features.	Reassess the affected requirements and modify or remove requirements that cannot realistically be implemented.	A Jara	Open
R05	The project team may underestimate the technical, economic, operational, schedule or legal feasibility of the proposed system, which could result in an unrealistic project baseline.	Feasibility	2	3	6	Medium	Evaluate the technical, economic, operational, schedule and legal feasibility before establishing the project baseline.	Reassess the feasibility findings and adjust the scope, requirements or schedule to establish a realistic baseline.	A Jara	Open
R06	The project team may fail to identify important project risks, constraints and assumptions. Resulting in potential problems not being considered in the project plan.	Risk Management	2	3	6	Medium	Conduct a structured Phase 1 review covering project risks, constraints and assumptions before approving the baseline.	Conduct another risk review and update the risk register and project plan with the missing risks, constraints or assumptions.	A Jara	Open
R07	The system architecture and Supabase database design may not fully support the approved requirements, user roles and shared information required by both applications.	Architecture/Data base	2	3	6	Medium	Review the proposed architecture and Supabase database design against the approved requirements and user roles before development.	Review and modify the architecture or database design before development progresses further.	B Mbinga	Open
R08	The integration between the Flutter application, ASP.NET web application and Supabase may not function as required, which could result in information not being shared correctly between the platforms.	Integration/ Testing	3	3	9	High	Define integration requirements early and perform integration checks to confirm that Flutter and ASP.NET can use the shared Supabase backend correctly.	Investigate the integration problem, prioritise the core data sharing functionality and correct the affected components.	TV Thabisi	Open
R09	The Flutter mobile application may not implement the approved requirements correctly, which could result in missing or incorrectly functioning mobile features.	Development /Flutter	2	3	6	Medium	The developers should develop the Flutter application against the approved SRS and review each implemented feature against the agreed requirements.	Review the affected Flutter functionality against the approved requirements and correct or complete the affected features.	ML Mwenda	Open

Feasibility and Risk Register Review

Finding	Does the project need to change ?	What change is required ?
Technical Feasibility	No	
Operational Feasibility	Yes	Schedule regular stakeholder feedback meetings throughout development.
Economic Feasibility	Yes	Utilise university free data where available. No major cost changes required.
Schedule Feasibility	yes	Prioritise MVP features, Postpone advanced analytics to stretch goals. Remove features that cannot be completed within 9 weeks.
Legal Feasibility	yes	Include consent forms, data deletion, and data access functionality to ensure POPIA compliance.
Ethical Feasibility	yes	Thoroughly test emergency panic button and SOS features. If testing reveals issues, simplify or provide fallback (display crisis line number)
High Project risks	yes	Remove or postpone unnecessary features, review each feature against requirements

Refined the Project Scope

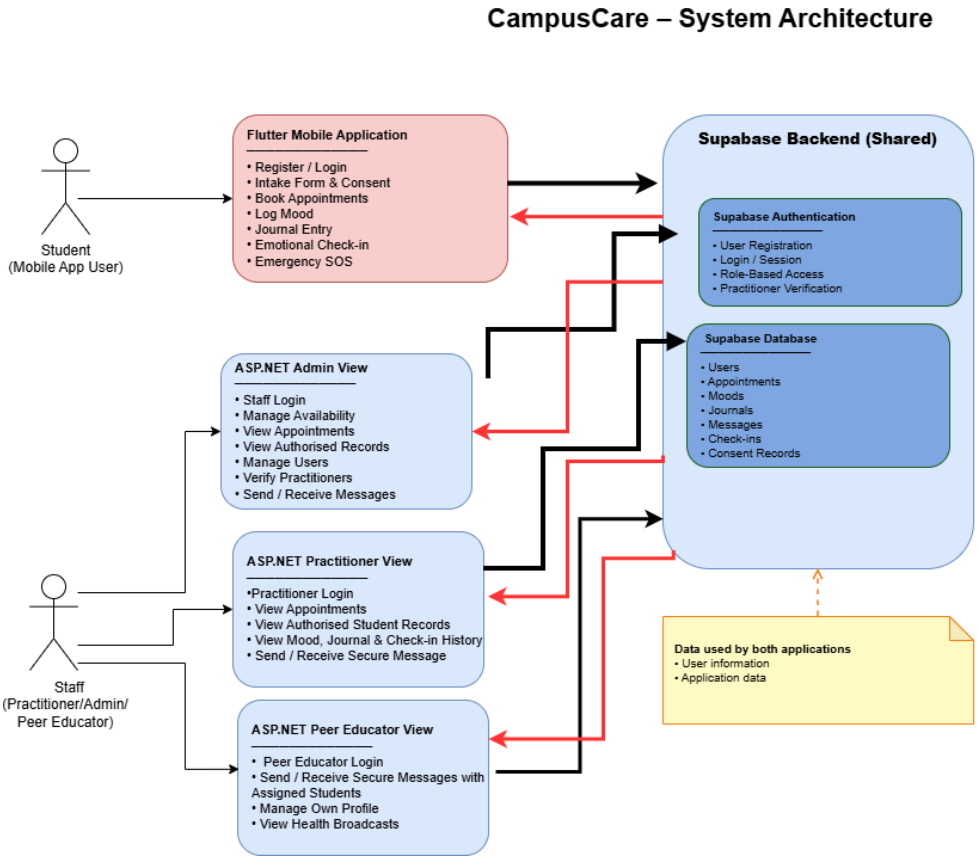
Function / Requirement	Decision	Reason	Stakeholder Informed ?
Register and login	Keep	Essential for system access	Yes
Digital Wellness Centre Intake Form	Keep	Core requirement for digitising manual process	Yes
Next-of-kin information	Keep	Required for emergency contact	Yes
Purpose of visit selection	Keep	Essential for triage and reporting	Yes
Student mood tracking	Keep	Core wellness feature	Yes
Student journaling	Keep	Core wellness feature	Yes
Student emotional check-ins	Keep	Core wellness feature	Yes
Secure messaging	postponed	Stakeholder requirement for student-provider communication	Yes
Emergency SOS / Panic button	Keep	Critical safety feature	Yes
Advanced reporting	Simplify	Basic reports only; detailed analytics postponed	Yes
SI/Tutor integration	Postpone	Requires additional coordination and scope expansion	Pending

3. Updated the Requirements

Requirement ID	Original requirement	Change made	Reason for change
FR14	The student should be able to use medication reminders	Simplified to basic push notifications	Reduces development complexity
FR15	The system should provide streaks and achievements	pending	Additional development time required
FR16	The system should provide additional reporting	Simplified to basic appointment reports	Advanced analytics postponed due to time constraints

Design of System Architecture

System Architecture Diagram



System Architecture Design Source Link :
https://drive.google.com/file/d/1T3yudGf6AKdJ6OFCDsgc7J9ftjGHCgY4/view?usp=sharing

Users of Each application

User/Role	Uses Flutter	Uses ASP.NET ?	Main purpose
Student	Yes	No	Uses the Flutter mobile app to access wellness support anytime and anywhere. Can book appointments, track moods, journal, and use emergency SOS without visiting the Wellness Centre physically.
Therapist	No	Yes	Uses the ASP.NET web app to manage their availability and view authorised student wellness records. monitor student progress between sessions.
Peer Educator	No	Yes	Uses the ASP.NET web app to manage their availability and student bookings. Can view limited student information and provide secure peer support to assigned students
Administrator	No	Yes	Assists with student intake and coordinates appointment flow at the Wellness Centre.

Responsibility of Each Component

Component	Purpose	Main Responsibilities
Flutter Mobile Application	Mobile app for the students	registration/login, intake form, consent, appointment booking, mood tracking, journaling, emotional check-ins, , wellness resources, emergency SOS, profile management.
ASP.NET Web Application	Web dashboard for therapist ,peer educators and administrators to view data and reports	Therapist and peer educator availability management , view authorised student records
Supabase Authentication	Manages secure login, registration, role-based access, and user verification for both applications	Manages secure registration, login, role-based access control, session handling, and practitioner verification status across both Flutter and ASP.NET.
Supabase Database	Shared backend database storing all system data	Shared backend database storing all system data – users, appointments, moods, journals, check-ins, messages, consent records – accessible by both applications where authorised.

Data Flow Between Components

Student → Flutter → Supabase (data captured and stored)

Supabase → ASP.NET → Staff (data retrieved and actioned)

Staff → ASP.NET → Supabase → Flutter (updates synced back to student)

From	To	Information Exchanged
Student	Flutter	Registration details, login credentials, intake form data, consent, next-of-kin information, purpose of visit, appointment requests, mood entries, journal entries, emotional check-ins, medication reminders, SOS alerts, profile updates
Flutter	Supabase	All student-submitted data (registration, intake, appointments, moods, journals, check-ins, messages, SOS) is stored in the shared Supabase database and authenticated via Supabase Authentication
Supabase	ASP.NET	Student appointment requests, mood logs, journal entries, check-ins, intake/consent records
ASP.ET	Staff	Appointment schedules, student wellness records (authorised), secure messages, reports, and verification status are presented to staff for action.
Staff	ASP.NET	Availability updates, appointment confirmations/cancellations, session notes (where applicable), secure messages, practitioner verification decisions, and report requests.
ASP.ET	Supabase	Staff-submitted data (availability, appointment updates, messages, verification status, reports) is stored in the shared Supabase database.
Supabase	Flutter	Updated appointment confirmations, practitioner availability, secure messages, wellness resources, and profile/verification status are synced back to the student's mobile app

Explanation of the architecture

Who Uses the Flutter Application

The Flutter mobile application is used only by students. It allows them to register, log in, complete intake forms, book appointments, log moods, journal, complete emotional check-ins, and use emergency SOS.

Who Uses the ASP.NET Application

The ASP.NET web application is used by administrators, therapists/practitioners, and peer educators. It allows them to manage users, verify practitioners, manage appointments and availability, view authorised student records, and send/receive secure messages.

What Supabase is Responsible For

Supabase is the shared backend responsible for authentication and database services. It manages secure login, session handling, and role-based access for all users. It also stores all system data such as users, appointments, moods, journals, messages, and check-ins in one central database.

How Flutter Communicates with Supabase

The Flutter app communicates with Supabase over secure HTTPS connections. It sends student login credentials to Supabase Authentication and reads/writes data (appointments, moods, journals, check-ins) to the Supabase Database. Supabase then returns confirmations, messages, and updated information back to the app.

How ASP.NET Communicates with Supabase

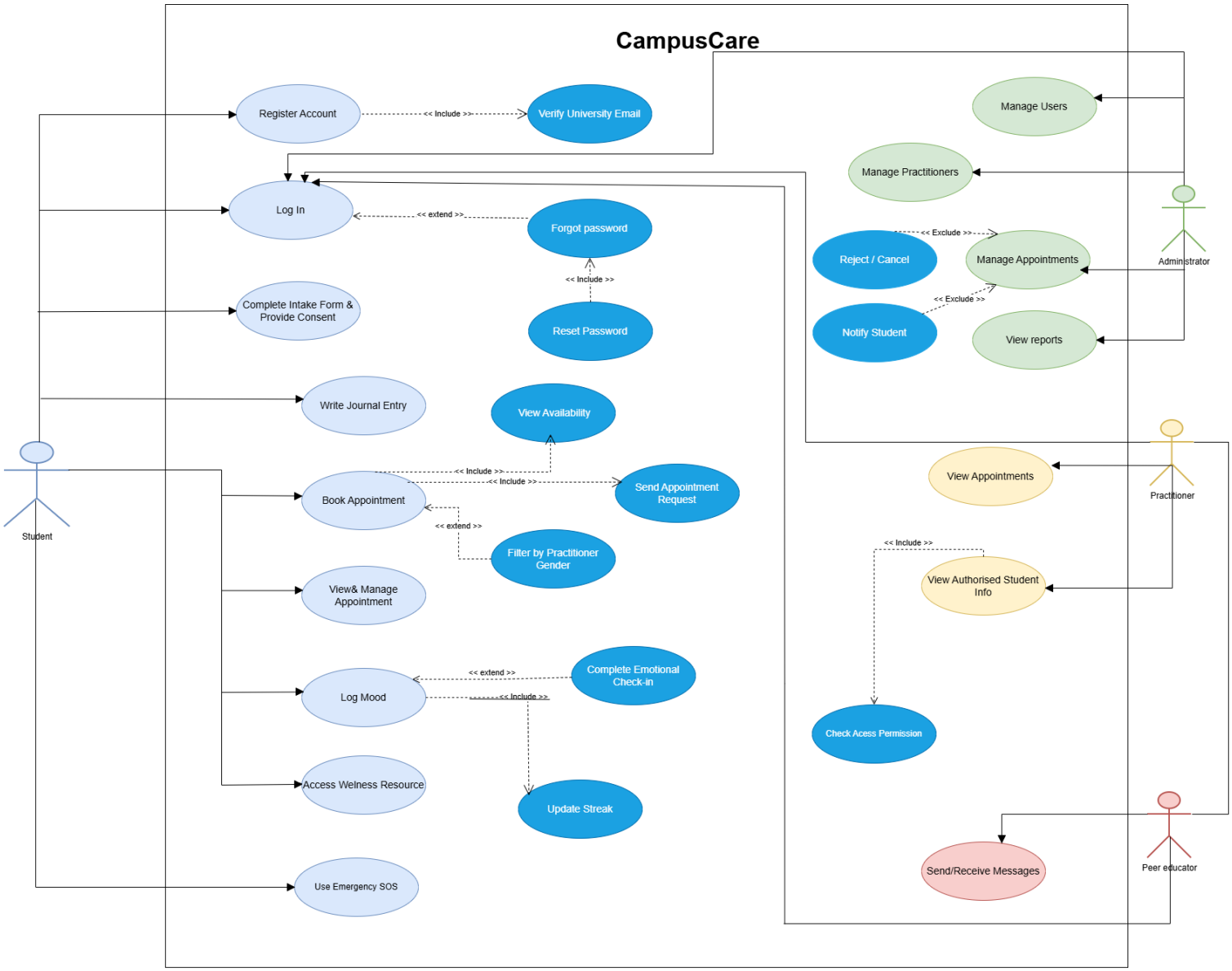
The ASP.NET web app communicates with Supabase in the same way, using secure HTTPS connections. It sends staff login credentials to Supabase Authentication and reads/writes data (availability, appointments, messages, verification decisions) to the Supabase Database. Supabase returns authorised student records and system data back to the web dashboard.

How the Two Applications Form One Integrated Solution

Flutter and ASP.NET are two separate frontends that share the same Supabase backend. When a student books an appointment on Flutter, it is stored in Supabase and appears instantly on the therapist's ASP.NET dashboard. When a therapist updates their availability on ASP.NET, it is stored in Supabase and reflected on the student's Flutter app. This shared backend ensures real-time synchronisation, unified authentication, and consistent data across both applications, forming one integrated CampusCare system.

The UML and Database Design

Use Case Diagram

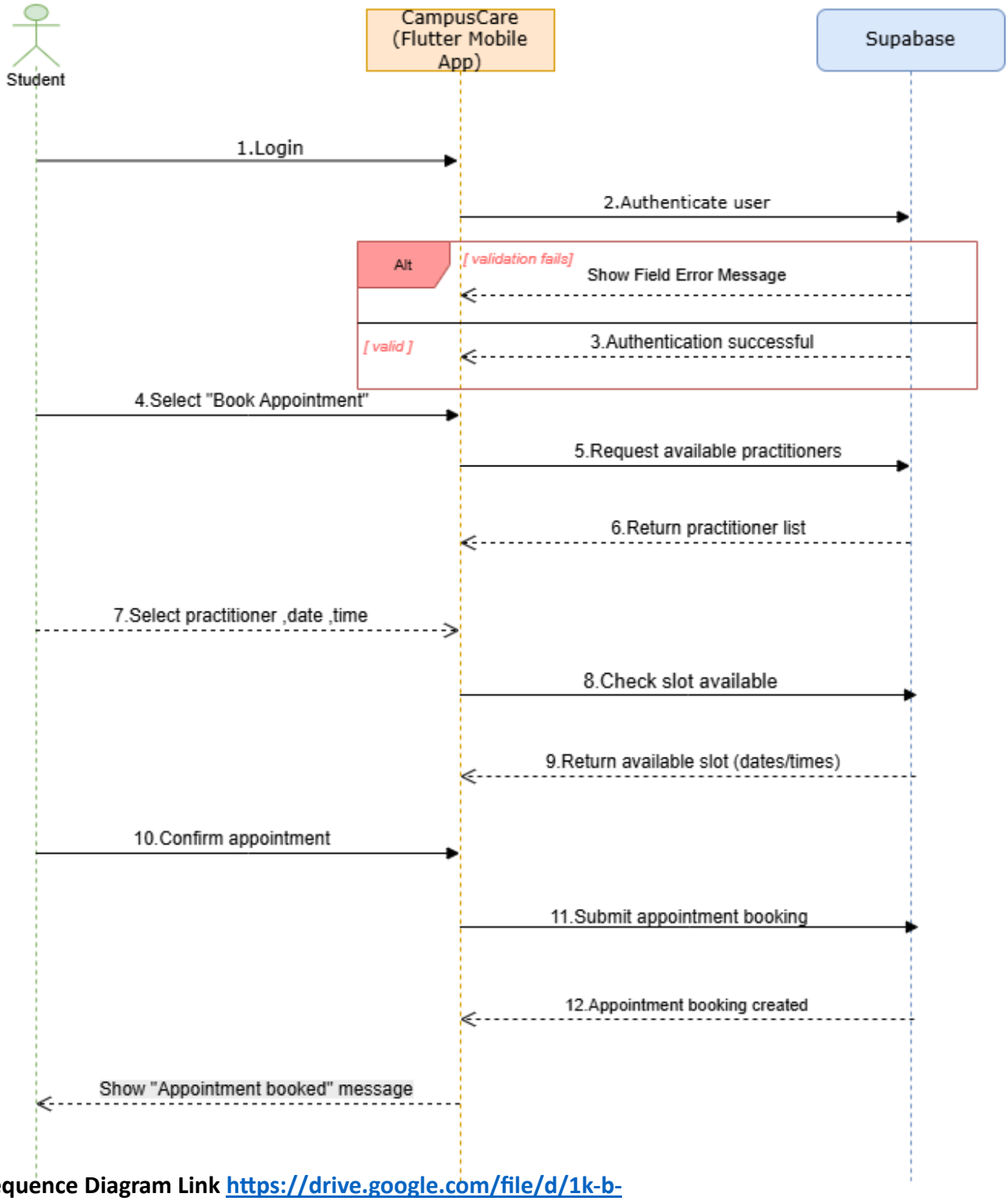


Use Case Diagram Link:
https://drive.google.com/file/d/1OY4BUxgNfjdjOU1JQ1PecuAbO6-OyIkn/view?usp=sharing

Sequence Diagrams

Flutter mobile application Sequence diagram

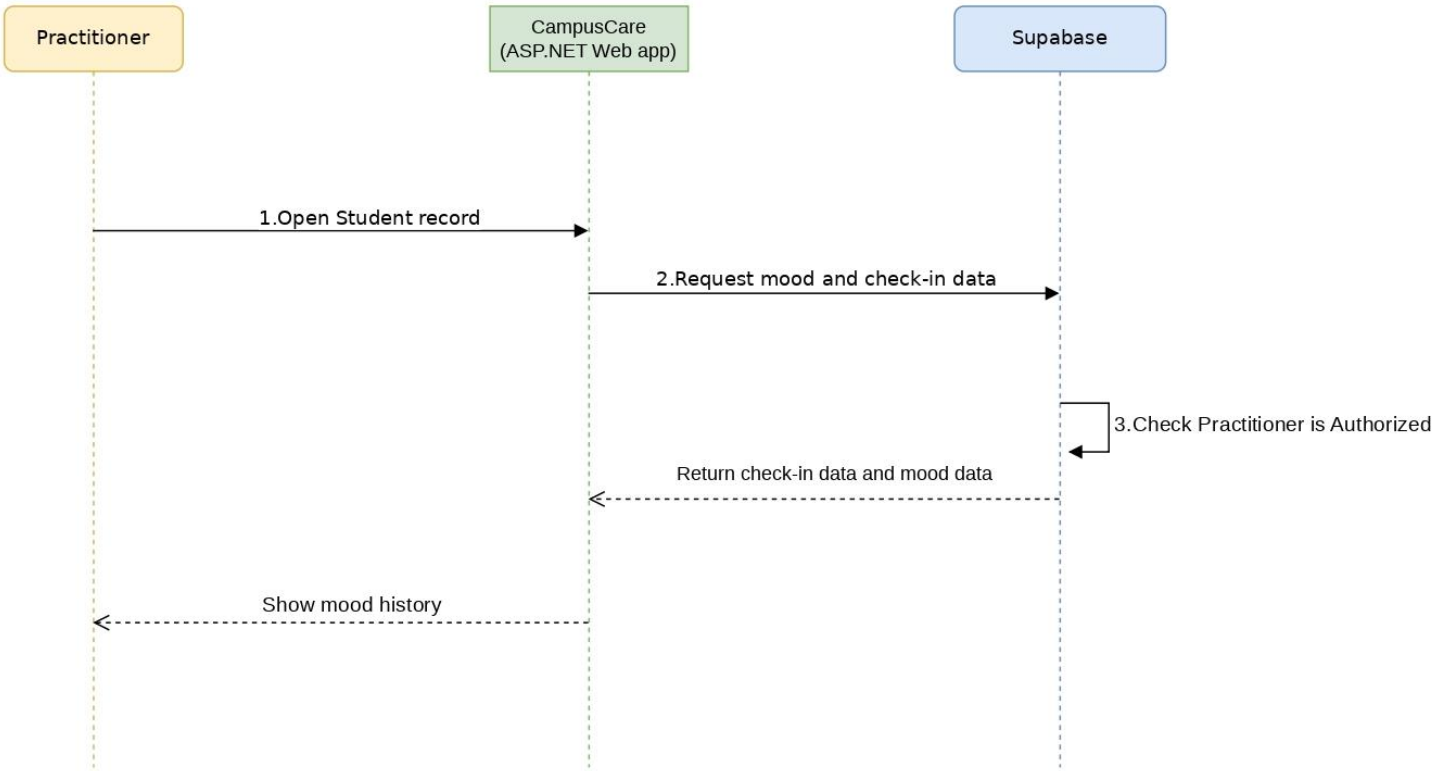
Booking appointment Sequence



Sequence Diagram Link https://drive.google.com/file/d/1k-b-WpGeeXKNERixGbxc11GlBPgPcklR/view?usp=sharing

ASP.NET web application sequence

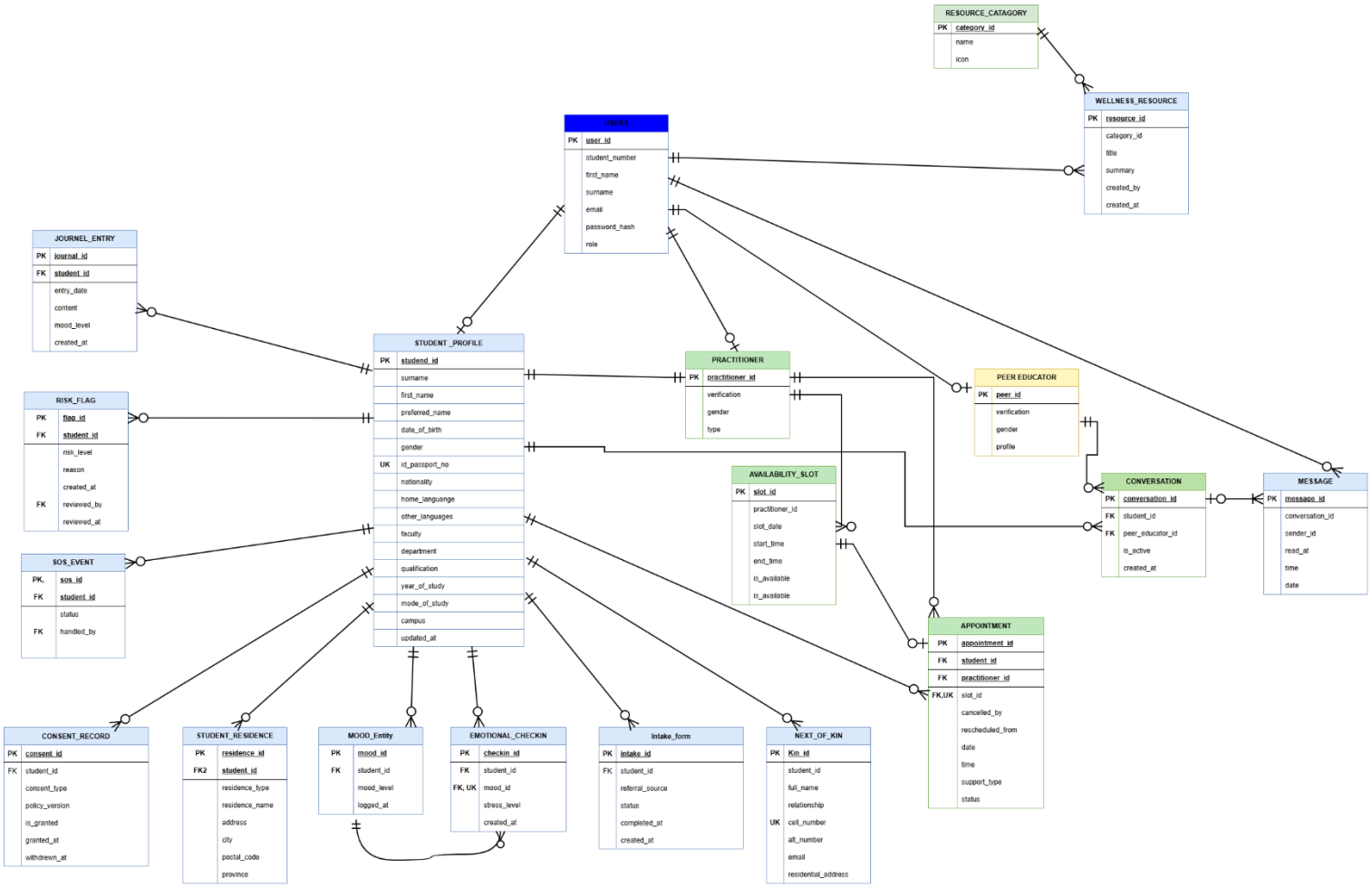
Mood check-in ,Streak update ,Authorized practitioner view Sequence



Sequence Diagram Link:

https://drive.google.com/file/d/1e1NmwJMd6LpJY7bJgYHu8NBqIcdodth7/view?usp=sharing

ERD Diagram



ERD Diagram Link:

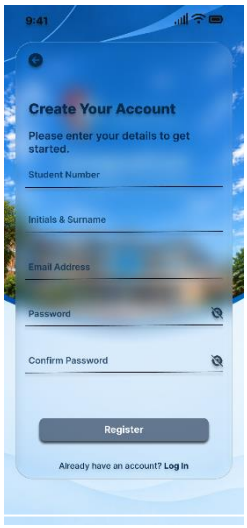
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Designs of the User Interfaces and the System Design

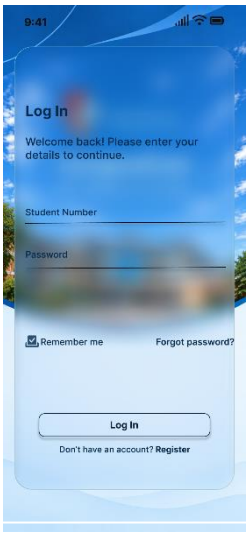
Required Flutter Screens



Splash Screen



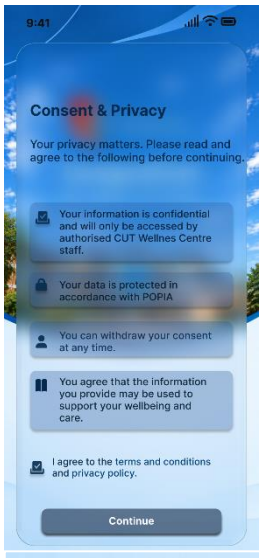
Registration Screen



Login Screen



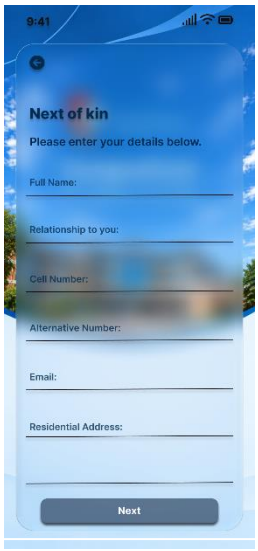
Forgot Password Screen



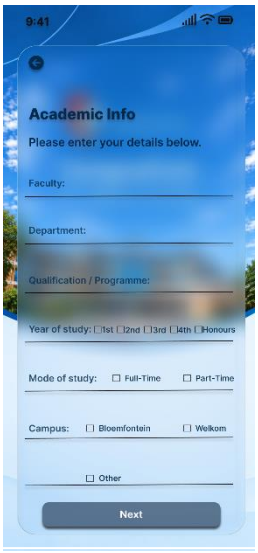
Consent Screen



Personal Info Screen



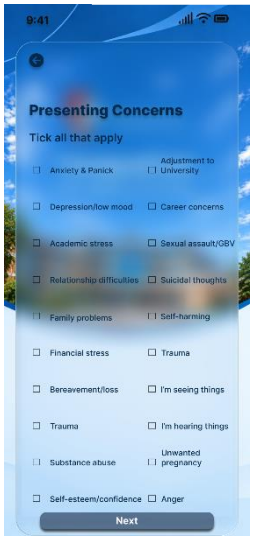
Next of Kin Screen



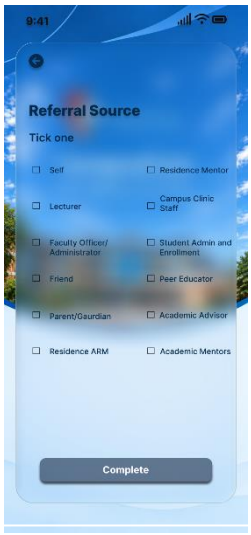
Academic Info Screen



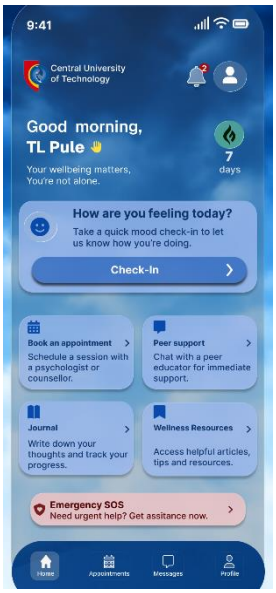
Residential Info Screen



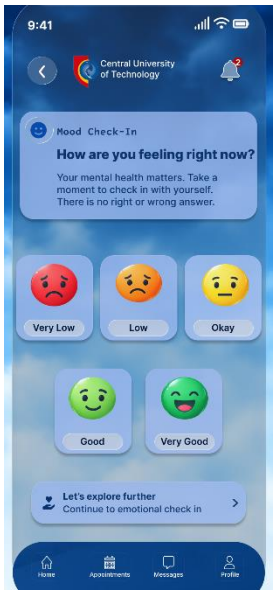
Presenting concerns Screen



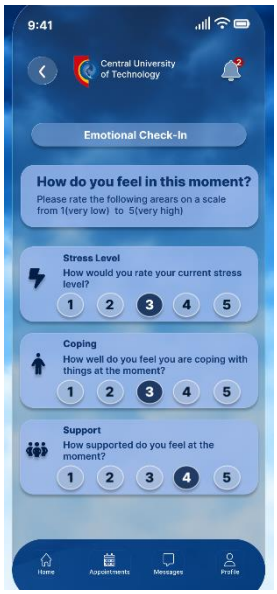
referral source screen



Home Dashboard Screen



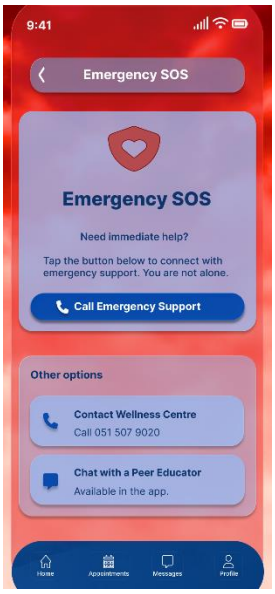
Mood check-in Screen



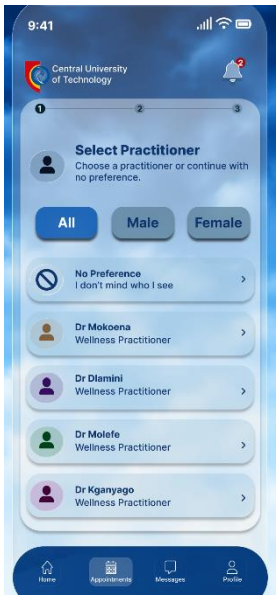
Emotional check-in Screen



Streak Screen



Emergency Screen



All Practitioners Screen



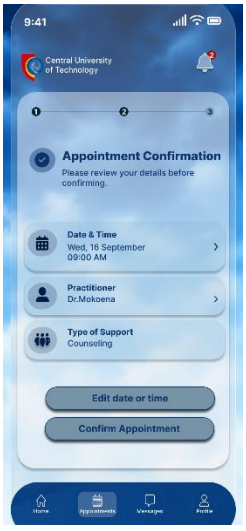
Male Practitioners Screen



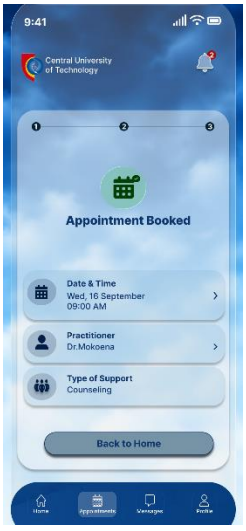
Female Practitioners Screen



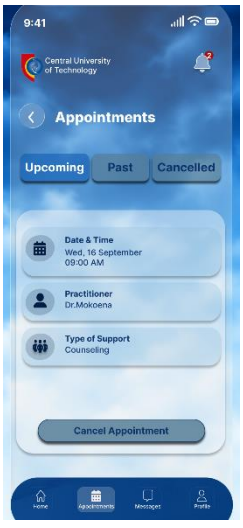
Book Appt. Screen



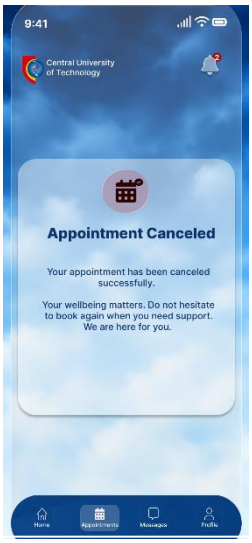
Edit Appt. Screen



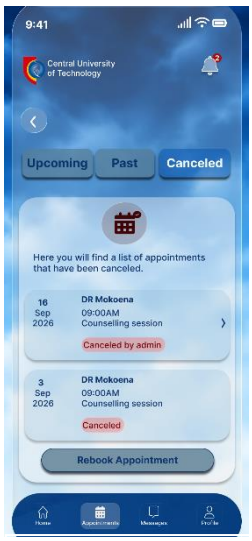
Appt. Booked Screen



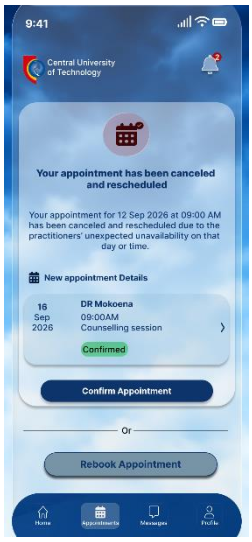
Upcoming Appt. Screen



Appt. cancelled



Cancel Appt.



Confirm Appt. Screen



Past Appt. Screen



Journal Screen



Journal Screen



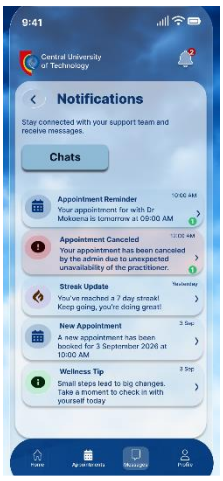
Profile Screen



Messages Screen



Message Screen



Notification Screen



Wellness Resources Screen

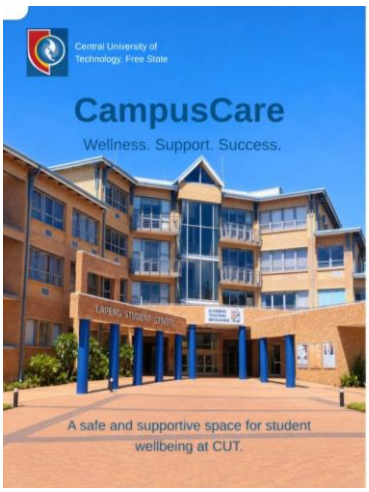
Figma Prototype/Wireframe Link:
https://www.figma.com/design/pIu7RB5B62NsLZreGgQjyV/CanpusCare?node-id=0-1&t=T8DZQBrBHjEe0oVQ-1

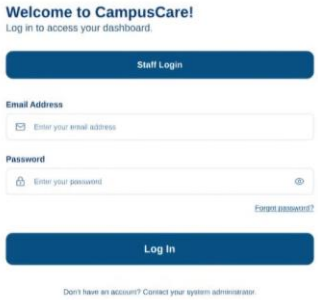
Navigation between screens

1. Splash → Registration → Consent → Personal Info → Next of Kin → Academic Info → Residential Info → Presenting Concerns → Home Dashboard
2. Splash → Login → Home Dashboard
3. Home Dashboard → Mood Check-in → Emotional Check-in → Home Dashboard
4. Home Dashboard → All Practitioners → Female Practitioners → Book Appt. → Confirm Appt. → Appt. Booked → Home Dashboard
5. Home Dashboard → Upcoming Appt. → Cancel Appt. → Appt. cancelled → Rebook → Book Appt.
6. Home Dashboard → Journal → New Entry → Save → Journal
7. Home Dashboard → Messages → Message Screen
8. Home Dashboard → Wellness Resources
9. Home Dashboard → Emergency SOS → Call / Chat
10. Home Dashboard → Profile → Personal Info / Intake / Consent

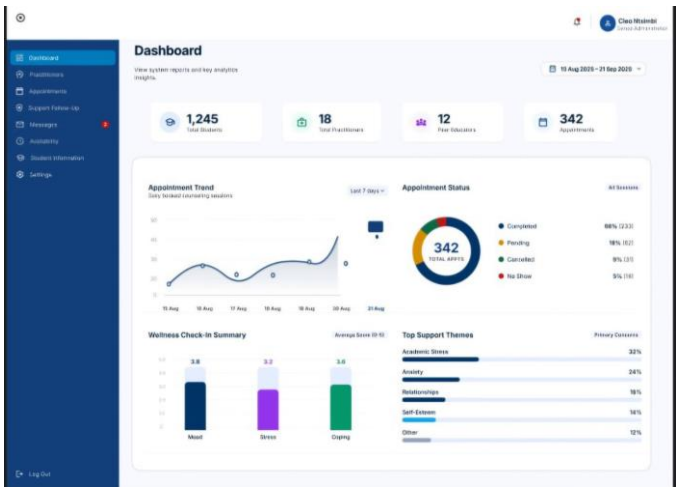
Required ASP.NET Screens

Figma Prototype/Wireframe Link:
https://www.figma.com/design/Q3R0ZAdLubL69gpXvT6jTD/Untitled?node-id=0-1&t=tg14ExU1fdjrHgx1-1

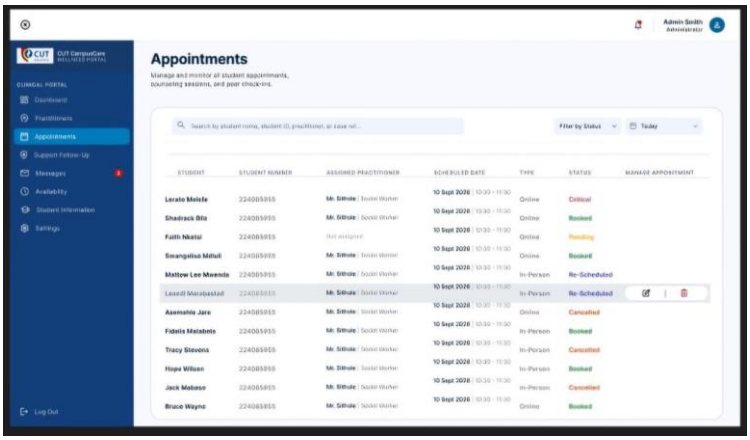




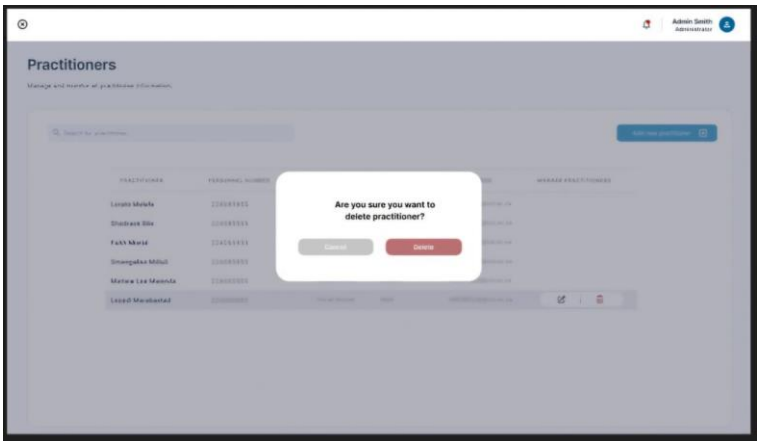
Admin Login Screen



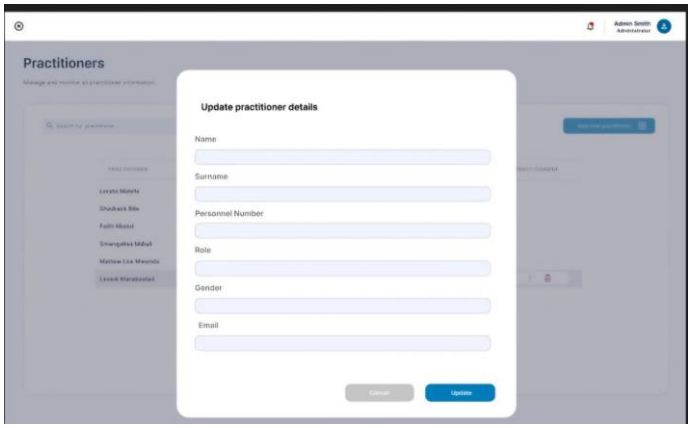
Admin Dashboard Screen



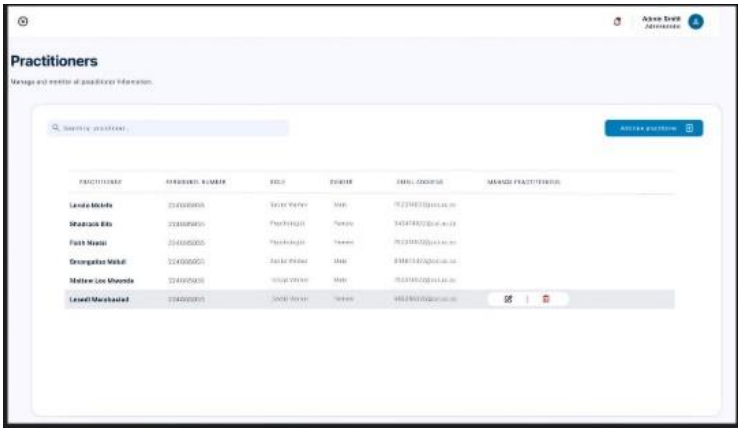
Admin Appt. Screen



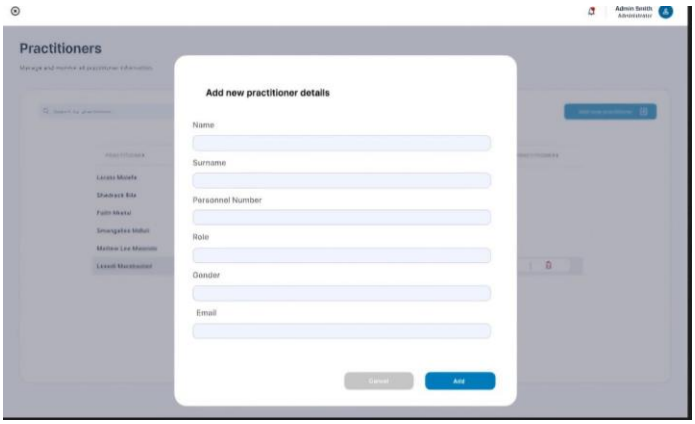
Confirm Deletion Screen



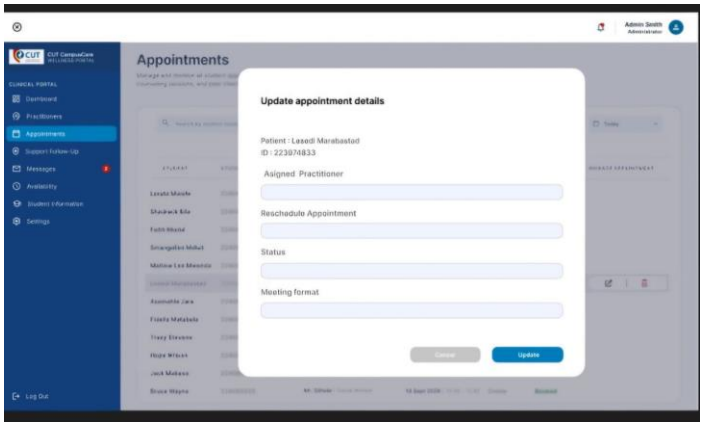
Admin update info Screen



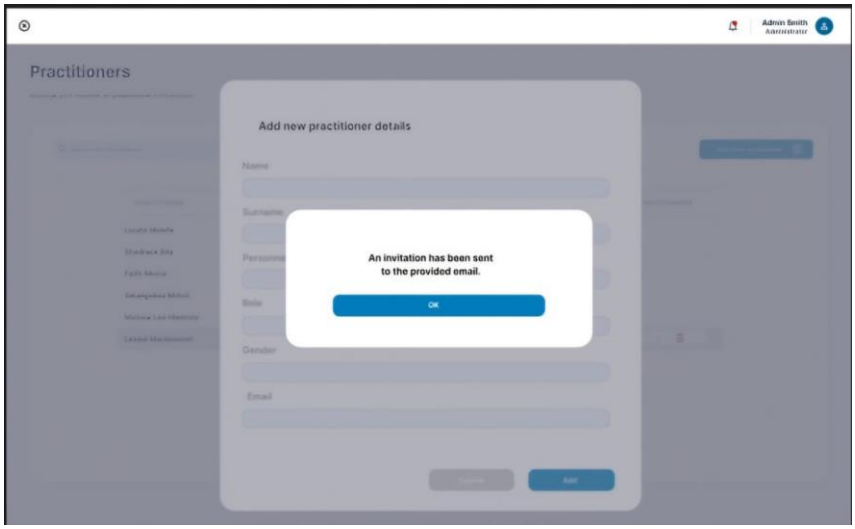
Practitioners' info Screen



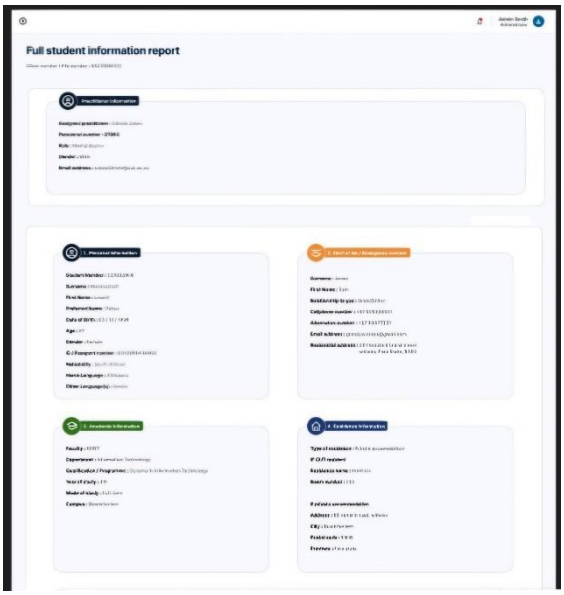
Admin Add info Screen



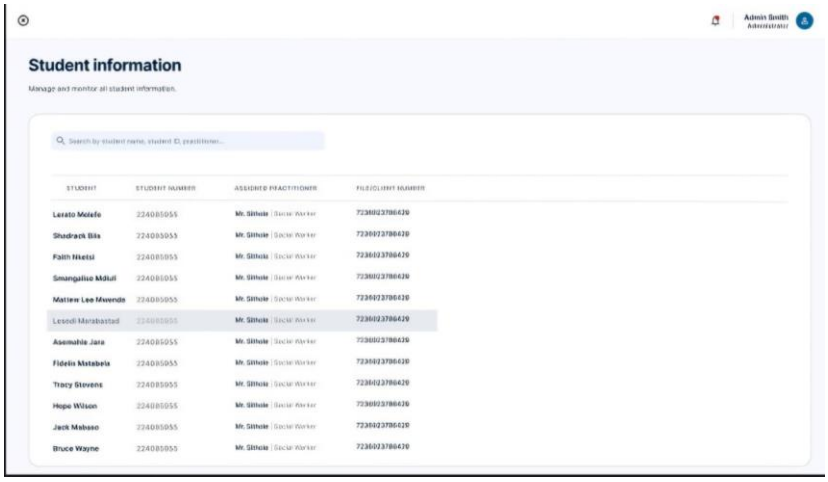
Admin update Appt. Screen



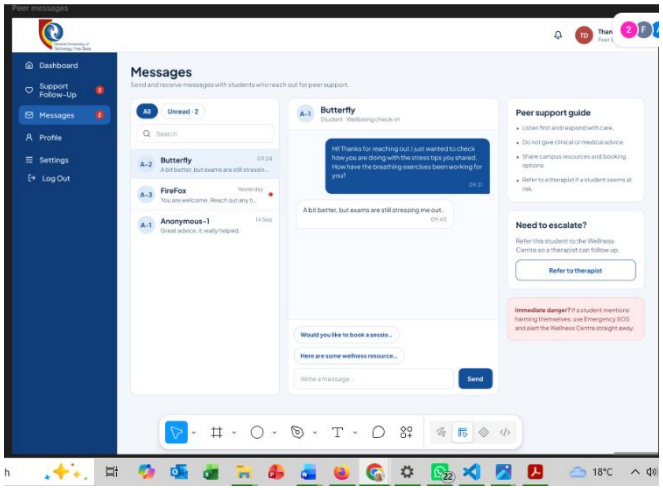
Invitation confirmation Screen



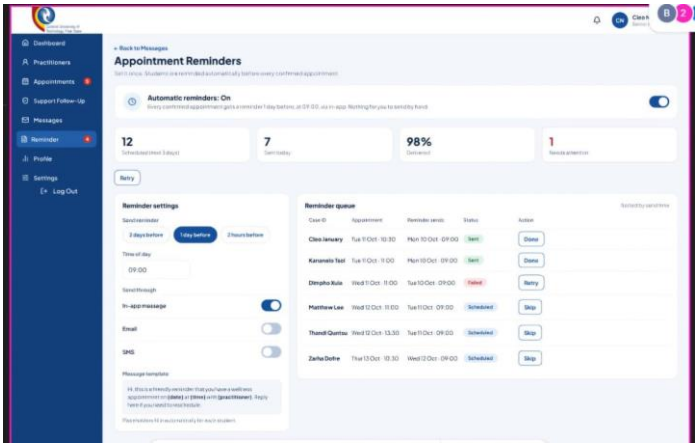
Student info Screen



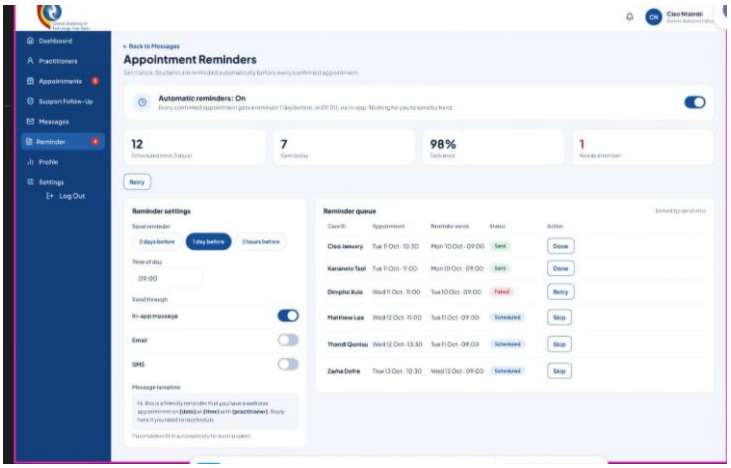
Admin Student Management Screen



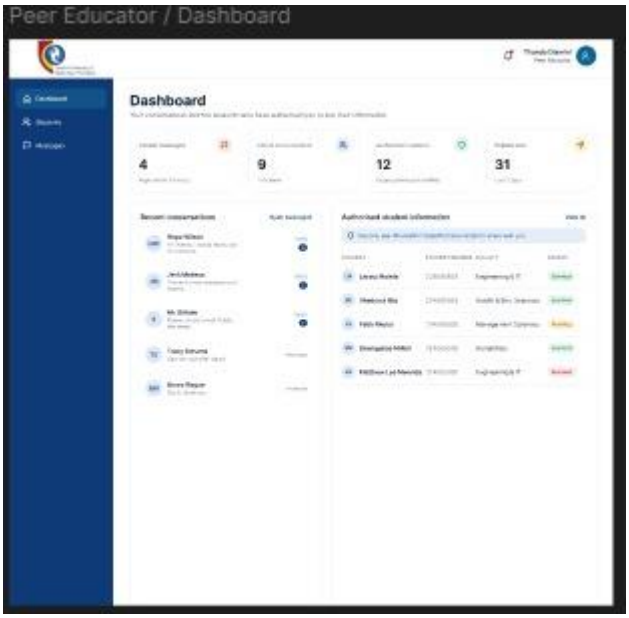
Message Dialogue Screen



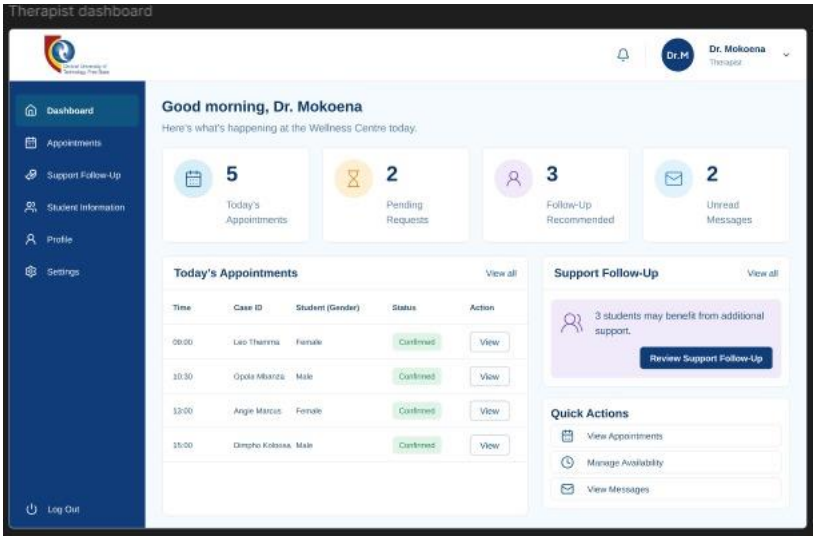
Appt. management Screen



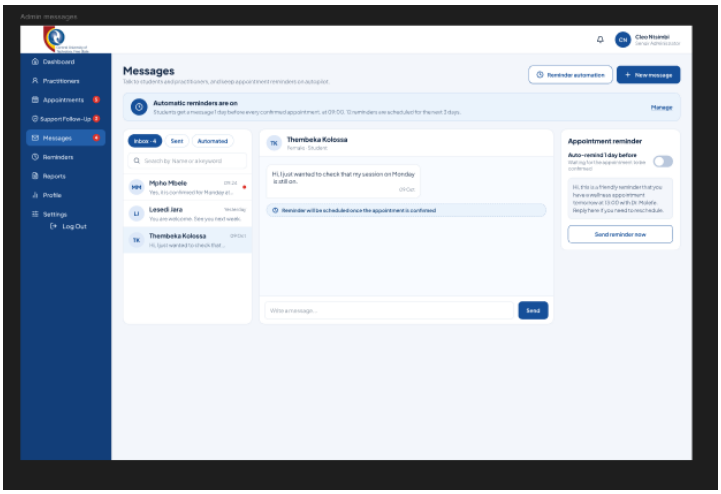
Appt. Reminder Screen



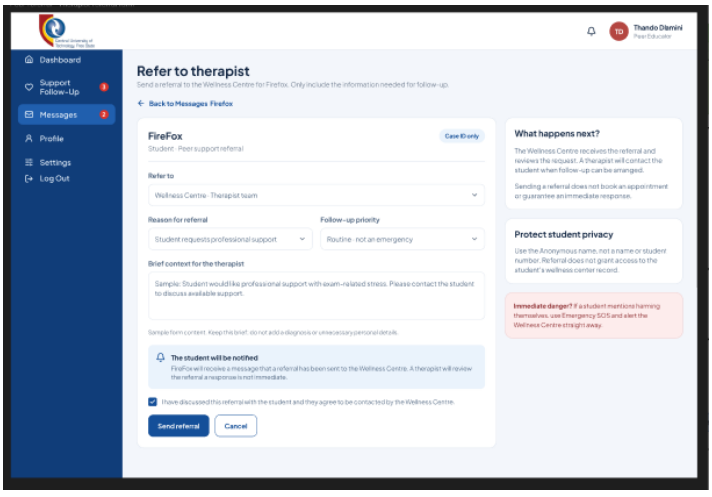
Peer Educator Dashboard Sreen



Practitioner Dashboard Sreen



Message Notification Screen



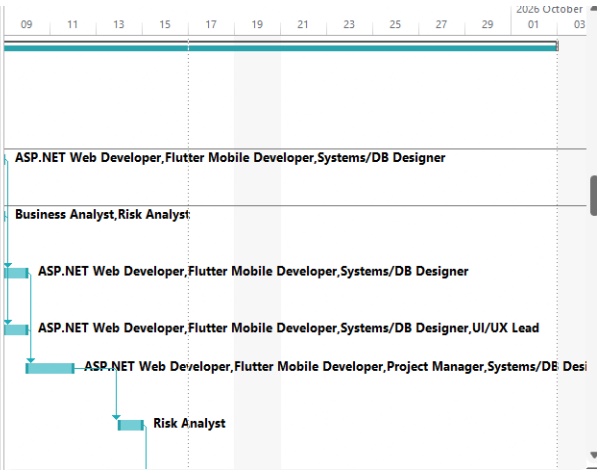
Referral Screen

Stakeholder Feedback

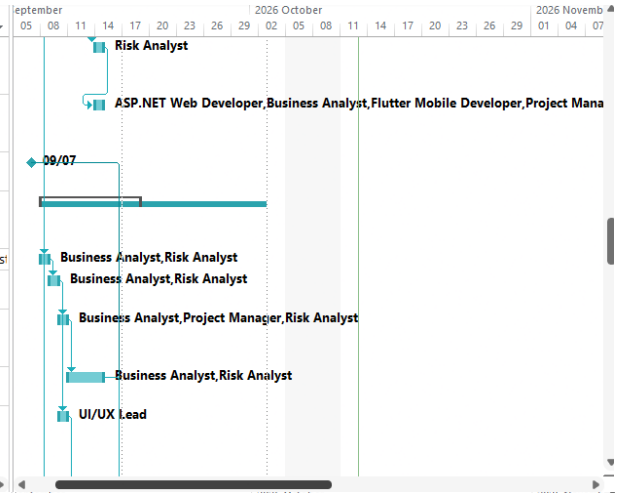
Item	Details
Date of feedback	2 October
Designs shown	Figma and ASAP.NET designs
Stakeholder comments	The stakeholder was impressed with the designs, but had a few changes to the prototypes
Changes requested	Logo change, colour adjustments, adding the preferred gender of practitioner, functions of the SOS, Removal of the medication reminder function
Changes made	Updated the logo, Added "Preferred Gender" filter (All / Male / Female) to the All-Practitioners Screen
Evidence available	Updated Figma prototype link, screenshot of updated Splash Screen, screenshot of updated Book Appointment Screen, screenshot of Emergency SOS Screen, Teams meeting notes (2 Oct 2026), WhatsApp confirmation from stakeholder

Updated Microsoft project plan

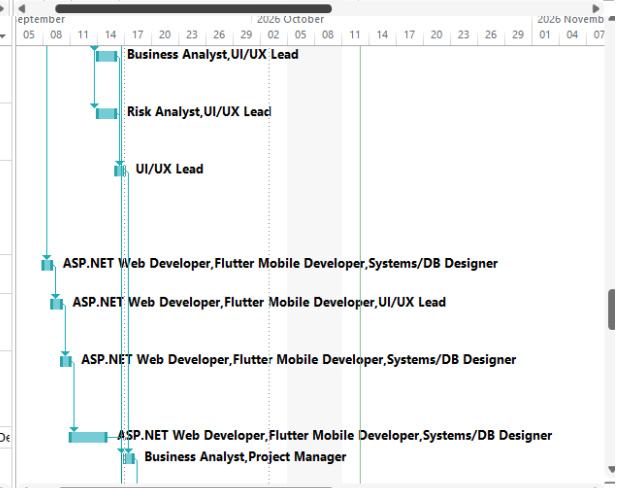
	Task Mode	Task Name	Duration	Start	Finish	Predecessor	Resource Names
28		**Phase 2: System Architecture and Design**	**20 days?**	**Mon 26/09/07**	**Fri 26/10/02**		
29		Review Phase 1 requirements and SRS	1 day	Mon 26/09/07	Mon 26/09/07		
30		Identify main system components	1 day	Tue 26/09/08	Tue 26/09/08	29	ASP.NET Web Developer Flutter Mobile
31		Identify users/roles for each application	1 day	Tue 26/09/08	Tue 26/09/08	29	Business Analyst Risk Analyst
32		Define responsibilities of each components	1 day	Wed 26/09/09	Wed 26/09/09	30	ASP.NET Web Developer Flutter Mobile
33		Identify main data flows	1 day	Wed 26/09/09	Wed 26/09/09	31	ASP.NET Web Developer
34		Draw system architecture diagram	2 days	Thu 26/09/10	Fri 26/09/11	32,33	ASP.NET Web Developer Flutter Mobile
35		Write architecture explanation	1 day	Mon 26/09/14	Mon 26/09/14	34	Risk Analyst



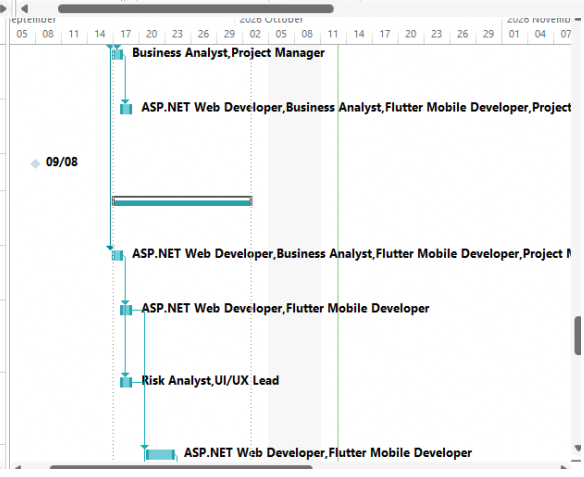
	Task Mode	Task Name	Duration	Start	Finish	Predecessor	Resource Names
35		Write architecture explanation	1 day	Mon 26/09/14	Mon 26/09/14	34	Risk Analyst
36		Review architecture with group	1 day	Mon 26/09/14	Mon 26/09/14	35	ASP.NET Web Developer Business
37		Architecture Complete	0 days	Mon 26/09/07	Mon 26/09/07		
38		**Develop the UML and Database Design**	**9 days?**	**Tue 26/09/08**	**Fri 26/09/18**		
39		Identify actors	1 day	Tue 26/09/08	Tue 26/09/08	29	Business Analyst
40		Identify use cases	1 day	Wed 26/09/09	Wed 26/09/09	39	Business Analyst
41		Determine use case relationships	1 day	Thu 26/09/10	Thu 26/09/10	40	Business Analyst Project
42		Draw Use Case Diagram	2 days	Fri 26/09/11	Mon 26/09/14	41	Business Analyst
43		Select sequence diagram processes	1 day	Thu 26/09/10	Thu 26/09/10	40	UI/UX Lead



	Task Mode	Task Name	Duration	Start	Finish	Predecessor	Resource Names
45		Draw Sequence Diagram 1(Flutter)	2 days	Mon 26/09/14	Tue 26/09/15	44	Business Analyst UI/UX Lead
46		Draw Sequence Diagram 2(ASP.NET)	2 days	Mon 26/09/14	Tue 26/09/15	44	Risk Analyst, UI/UX Lead
47		Check sequence diagrams against use cases	1 day	Wed 26/09/16	Wed 26/09/16	45,46	UI/UX Lead
48		Identify entities	1 day	Tue 26/09/08	Tue 26/09/08	29	ASP.NET Web Developer
49		Identify attributes and keys	1 day	Wed 26/09/09	Wed 26/09/09	48	ASP.NET Web Developer Flutter Mobile
50		Determine relationships and cardinalities	1 day	Thu 26/09/10	Thu 26/09/10	49	ASP.NET Web Developer Flutter Mobile Developer
51		Draw ERD	2 days	Fri 26/09/11	Mon 26/09/1	50	ASP.NET Web De
52		Alignment check across	1 day	Thu 26/09/17	Thu 26/09/17	42,47,51	Business Analyst

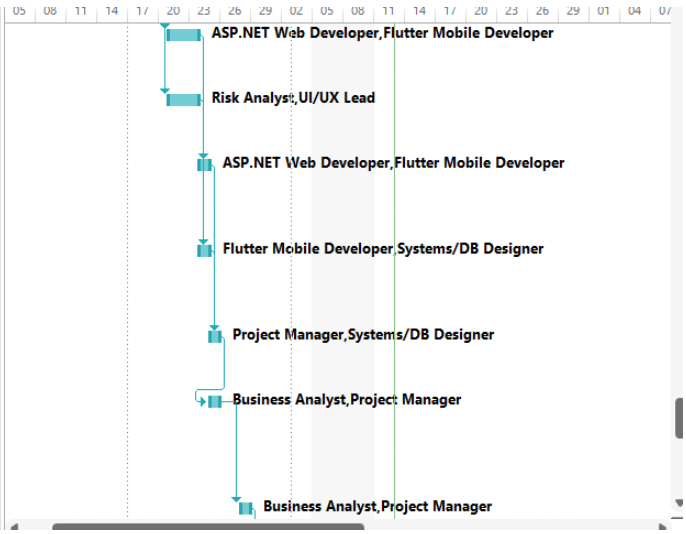


	Task Mode	Task Name	Duration	Start	Finish	Predecessor	Resource Names
52		Alignment check across UML and ERD	1 day	Thu 26/09/17	Thu 26/09/17	42,47,51	Business Analyst Project
53		Review UML and ERD with group	1 day	Fri 26/09/18	Fri 26/09/18	52	ASP.NET Web Developer Business
54		UML and ERD complete	0 days				
55		**Design the User Interfaces and Finalise**	**12 days**	**Thu 26/09/17**	**Fri 26/10/02**		
56		Review approved requirement	1 day	Thu 26/09/17	Thu 26/09/17	37	ASP.NET Web Developer Business
57		Identify required Flutter Screens	1 day	Fri 26/09/18	Fri 26/09/18	56	ASP.NET Web Developer Flutter Mobile Developer
58		Identify required ASP.NET pages	1 day	Fri 26/09/18	Fri 26/09/18	56	Risk Analyst, UI/UX Lead
59		Create	3 days	Mon	Wed	57	ASP.NET Web



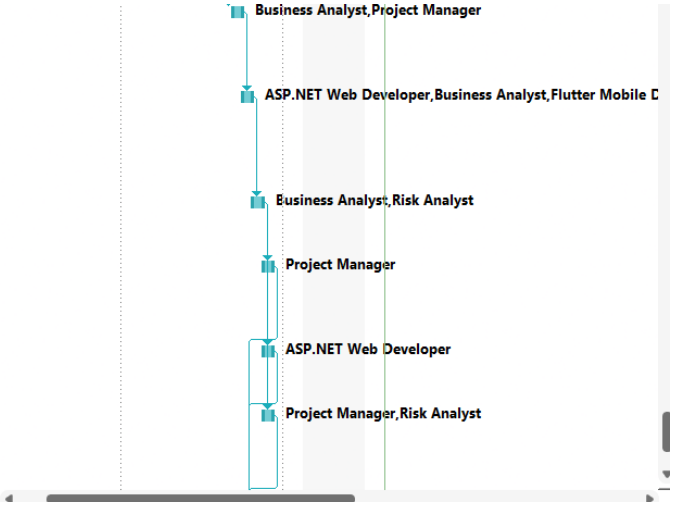
GANTT CHART

	Mode	Task Name	Duration	Start	Finish	Predecessor	Names
59		Create Flutter wireframes	3 days	Mon 26/09/21	Wed 26/09/23	57	ASP.NET Web Developer Flutter Mobile
60		Create ASP.NET wireframes	3 days	Mon 26/09/21	Wed 26/09/23	58	Risk Analyst, UI/UX Lead
61		Show navigation between screens	1 day	Thu 26/09/24	Thu 26/09/24	59,60	ASP.NET Web Developer Flutter Mobile Developer
62		Check mobile and web integration	1 day	Thu 26/09/24	Thu 26/09/24	59,60	Flutter Mobile Developer Systems/DB Designer
63		Check design against ERD	1 day	Fri 26/09/25	Fri 26/09/25	61,62	Project Manager Systems/DB
64		Alignment check across complete system design	1 day	Fri 26/09/25	Fri 26/09/25	63	Business Analyst Project Manager
65		Obtain	1 day	Mon	Mon	64	Business



GANTT CHART

65		Obtain stakeholder feedback on interfaces	1 day	Mon 26/09/28	Mon 26/09/28	64	Business Analyst Project Manager
66		Update system design based on feedback	1 day	Tue 26/09/29	Tue 26/09/29	65	ASP.NET Web Developer Business Analyst Flutter Mobile
67		Update SRS with Phase 2 sections	1 day	Wed 26/09/30	Wed 26/09/30	66	Business Analyst Risk Analyst
68		Update Microsoft Project schedule	1 day	Thu 26/10/01	Thu 26/10/01	67	Project Manager
69		Update GitHub with design	1 day	Thu 26/10/01	Thu 26/10/01	67	ASP.NET Web Developer
70		Update Project Progress Tracker	1 day	Thu 26/10/01	Thu 26/10/01	67	Project Manager Risk Analyst



System Development and implementation Integration (Phase 3)

Understanding the Integrated System for CampusCare

Question	Our Answer
Which application will students use?	Students will use the Flutter mobile application.
Which application will Wellness Centre staff use?	Practitioners, Peer Educators, and Administra use the ASP.NET web application.
What is the purpose of Supabase?	Supabase acts as the shared backend infrastructure for both applications. It provides the central PostgreSQL database to store all records (users, appointments, moods, journals), handles secure user authentication and role-based access control
What happens when a student submits an appointment request?	When a student selects a practitioner and time slot on Flutter and confirms, the mobile app sends an API request (HTTPS) containing the booking details to Supabase. Supabase authenticates the student's session, saves the new appointment record in the database, and returns a success confirmation to the Flutter app.
Which CRUD operation is used to save a new appointment?	Create. The Flutter application performs a "Create" operation by inserting a new row of data into the Appointments table in the Supabase database.
Which CRUD operation is used when staff view appointment ?	Read. The ASP.NET web application performs a "Read" operation to query the Supabase database and retrieve the list of appointments, displaying them on the Therapist or Admin dashboard.
Which CRUD operation is used when a Admin cancels an appointment?	Delete, The ASP.NET web performs an 'Delete' operation to remove the students book then a notification is sent to the student that they will have to book an appointment for anther day.
Why must the two applications use the same backend?	They must use the same backend to ensure real-time synchronisation, data consistency, and a single source of truth. If they used separate databases, a student booking on mobile would never appear on the staff's web dashboard, resulting in duplicated bookings and system failure.
Where should the group maintain its source code?	The source code is maintained in a single GitHub repository

	(CampusCare_WIL_Group-F), organised into clear mobile/ (Flutter) and web/ ASP.NET)folders
How can members provide evidence of their development contributions?	Contributions are tracked via GitHub commit history, the ITC327W Project Progress Tracker

Set Up and Implement the shared Supabase Backend

Review of Our Approved Database Design

Entity / Table	Purpose	Main fields
User	Stores the core login credentials and basic identity the user of CampusCare	user_id (PK), student_number, first_name, surname, email, password_hash, role
Student_Profile	Stores the detailed, personal, and academic information collected during the student's initial intake process.	student_id ,surname, first_name, date_of_birth, gender, faculty, department, year_of_study, campus
PRACTITIONER	Stores specific details for Therapists/Psychologists, linking them to their user account and verifying their status.	practitioner_id ,verification, gender, type
PEER EDUCATOR	Stores specific details for student Peer Educators, including their verification status to ensure they are authorised to support others.	peer_id (PK/FK to User), verification, gender, profile
AVAILABILITY_SLOT	Represents the specific dates and times that a practitioner has opened up for students to book.	slot_id (PK), practitioner_id (FK), slot_date, start_time, end_time, is_available
APPOINTMENT	Links a student to a practitioner and an availability slot to formally book a counselling session.	appointment_id (PK), student_id (FK), practitioner_id (FK), slot_id (FK), cancelled_by, rescheduled_from, date, time, status
MOOD_ENTRY	Tracks the daily or regular emotional state of a student for self-monitoring and therapist visibility.	mood_id (PK), student_id (FK), mood_level, logged_at
EMOTIONAL_CHECKIN	Stores deeper, detailed check-ins where students rate their stress, coping mechanisms, and support levels.	checkin_id (PK), student_id (FK), mood_id (FK), stress_level, coping_level, support_level

JOURNAL_ENTRY	Stores private text entries written by the student for self-reflection, optionally linked to a mood.	journal_id (PK), student_id (FK), entry_date, content, mood_level, created_at
CONSENT_RECORD	Tracks whether a student has agreed to the POPIA privacy policy and terms, including when they consented or withdrew.	consent_id (PK), student_id (FK), consent_type, policy_version, is_granted, granted_at, withdrawn_at
INTAKE_FORM	Stores the specific clinical/referral information a student provides when seeking help (e.g., why they are visiting, who referred them).	intake_id (PK), student_id (FK), referral_source, status, completed_at
NEXT_OF_KIN	Stores emergency contact details for the student.	kin_id (PK), student_id (FK), full_name, relationship, cell_number, email
STUDENT_RESIDENCE	Tracks where the student is living (e.g., campus residence, private accommodation) which is useful for wellness context.	residence_id (PK), student_id (FK), residence_type, residence_name, address, city, postal_code
SOS_EVENT	Logs whenever a student presses the emergency panic button on the app, tracking who handled it.	sos_id (PK), student_id (FK), status, handled_by
RISK_FLAG	Automatically or manually flags a student as "at-risk" (e.g., if their mood logs are consistently low) so therapists can intervene.	flag_id (PK), student_id (FK), risk_level, reason, reviewed_by, reviewed_at
RESOURCE_CATEGORY	Groups wellness resources (e.g., Stress Management, Academic Support, Self-Care).	category_id (PK), name, icon

Preparations for Backend for Both Applications

Application	Information it will send or retrieve
Flutter	**Sends:** Student registration/consent data, new appointment requests, daily mood logs, emotional check-ins, journal entries **Retrieves:** Practitioner availability slots, appointment statuses (confirmed/cancelled), wellness resources, and replies from Peer Educators/Therapists.
ASP.NET	**Sends:** Practitioner availability updates, appointment confirmations/reschedules, and practitioner verification status. **Retrieves:** Student profile data, incoming appointment requests, student mood/ (if authorised), SOS alerts,

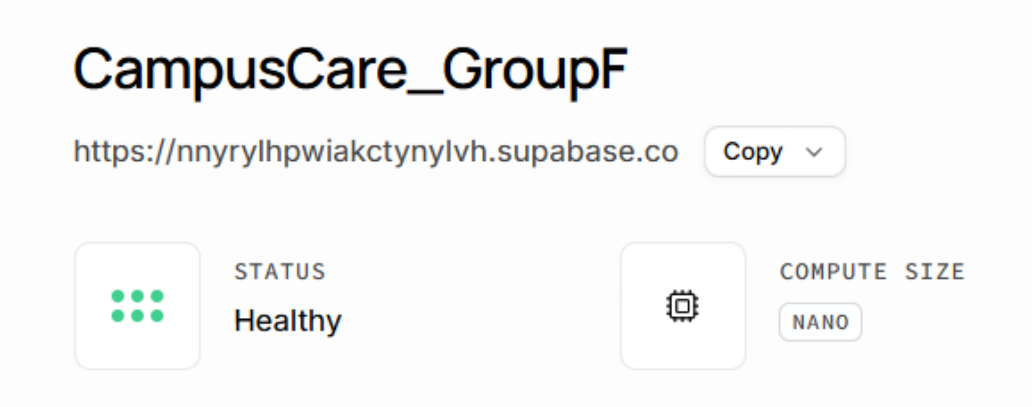
Initial Performed Backend Checks

Check	Result	Action required
Required tables have been created	**Yes, All core tables have been created in supa base**	**None**
Primary keys are configured	**Yes, All created tables have the Unique Primary key**	**None**
Foreign keys and relationships work	**Yes**	**None**
Test records can be added	**Yes, dummy records were inserted**	**none**
Test records can be retrieved	**Yess. Data was successfully queried and returned using the Supabase API and viewed in the Table Editor**	**None**
Records can be updated where required	**In progress, pending to confirmed status needs to be worked on**	**Complete the logic**
Authentication works, where required	**In Progress. Supabase Email/Password Authentication is enabled**	**Complete the authentication logic in the frontend apps and test user sessions**
Access permissions are appropriate	**In Progress. Row Level Security (RLS) is enabled. Basic rules are set (e.g., students can only read their own data).**	**Configure RLS policies specifically for Peer Educators and Therapists so they can only access data for students assigned to them.**

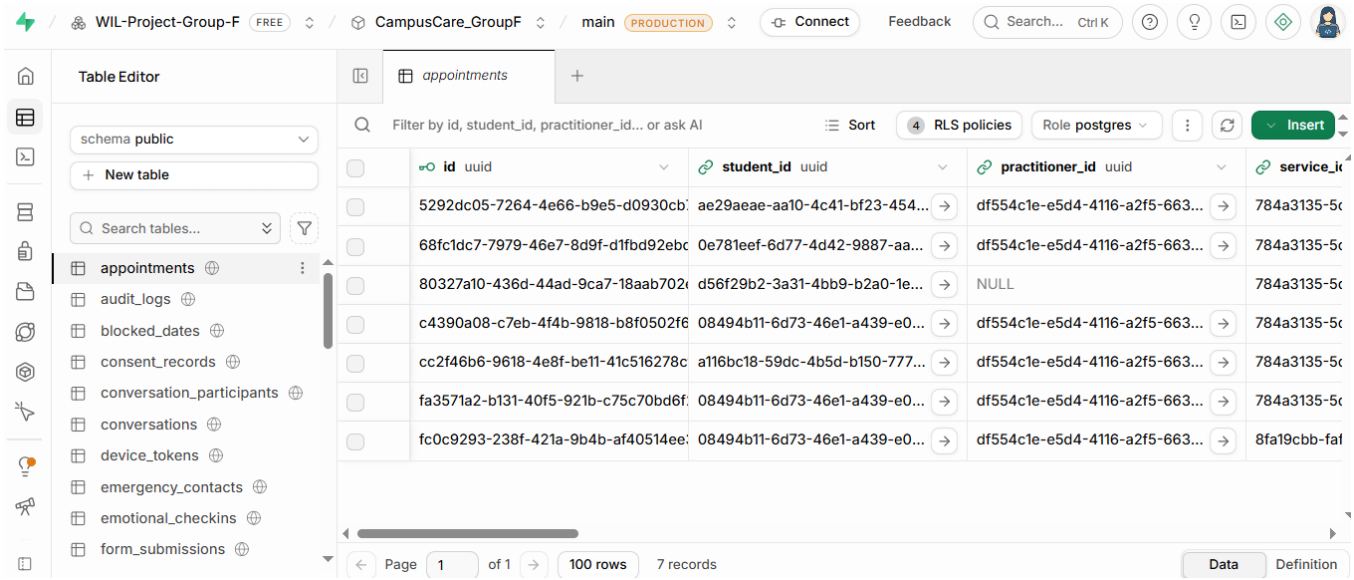
Evidence

Screenshot of the shared Supabase project

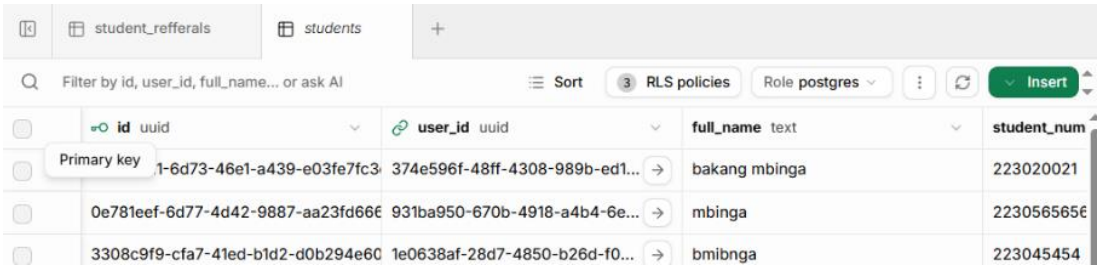
Supabase Link :https://nnyrylhpwiakctynylvh.supabase.co

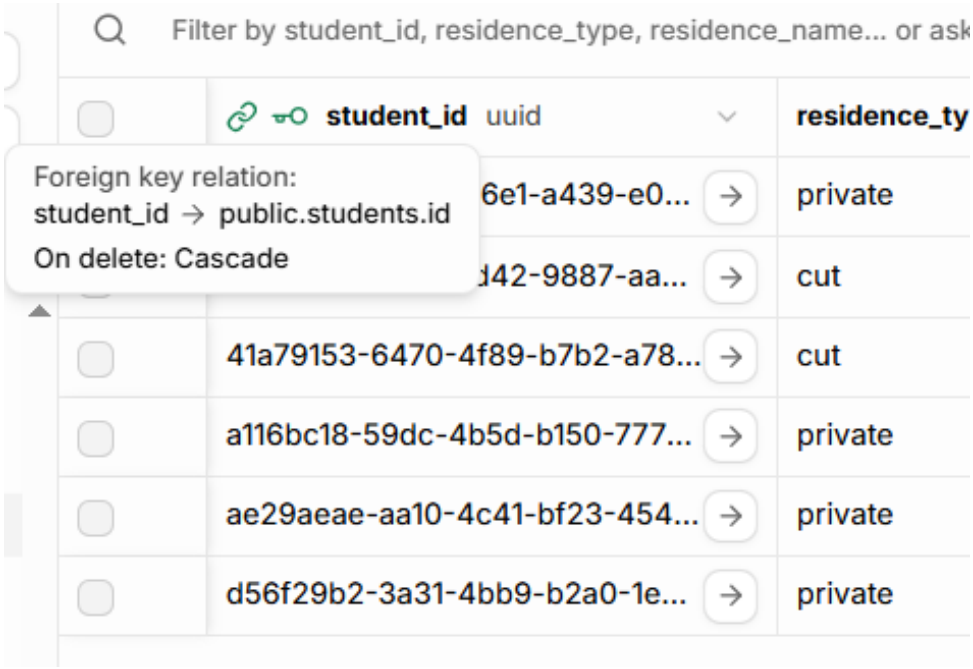


Screenshots of the implemented database tables

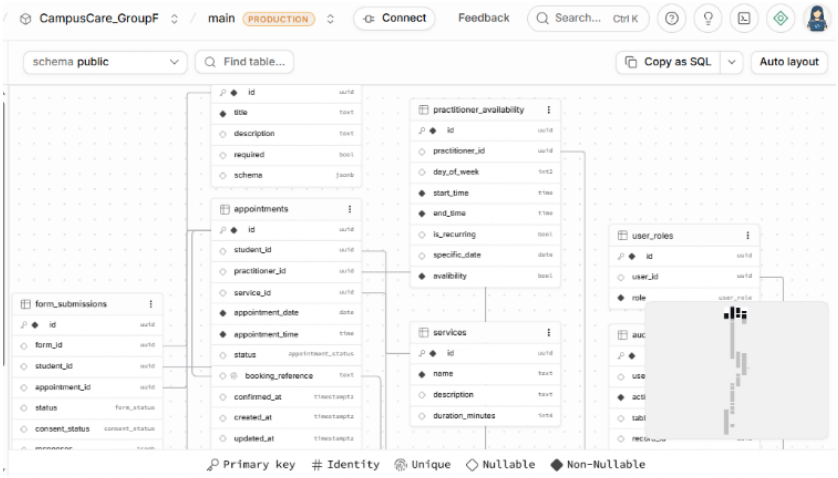


Evidence of primary keys and foreign keys

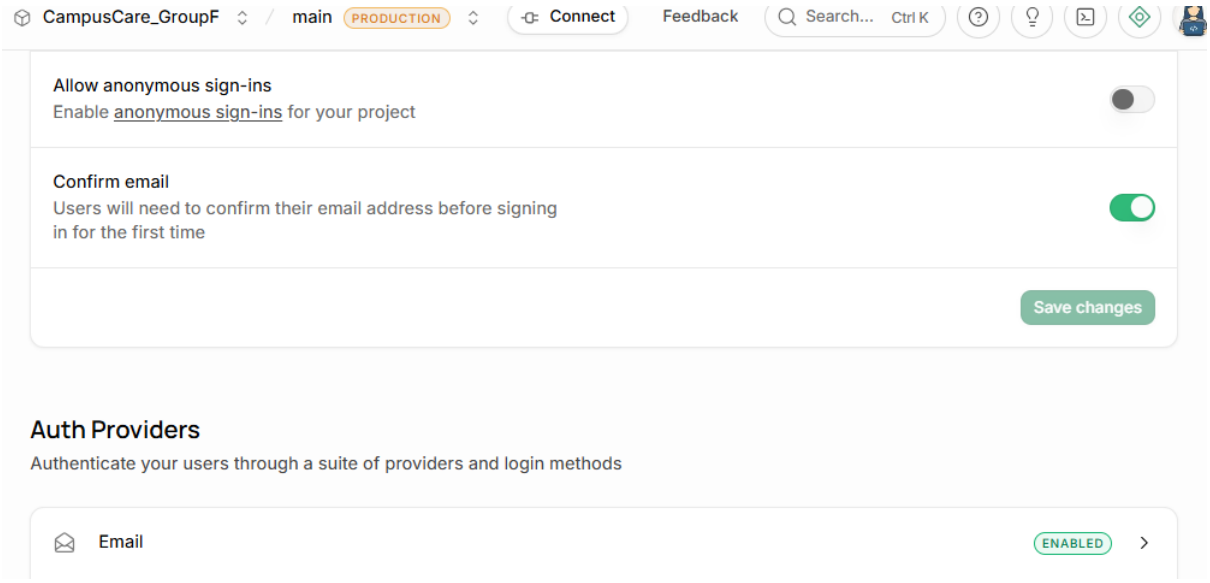




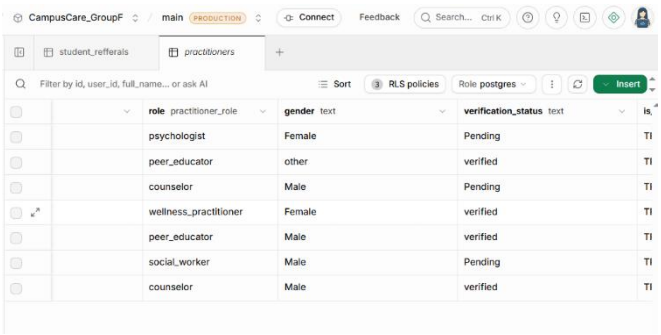
Evidence of relationships between tables



Authentication and access configuration, where required



Evidence of initial database checks using dummy data



Developing the Flutter Mobile Application

Review the Approved Mobile Application Requirements

Mobile function	Related requirement or use case	Required screen
User Registration & Login	FR1: The student must be able to register and log in.	Splash Screen, Registration Screen, Login Screen, Forgot Password Screen
Digital Intake & Consent	FR2, FR3, FR4, FR5: Complete intake form, provide next-of-kin, purpose of visit, and POPIA consent.	Consent Screen, Personal Info Screen, Next of Kin Screen, Academic Info Screen, Residential Info Screen, Presenting Concerns Screen
Manage Appointments	FR7: The student must be able to view and manage their appointments.	Upcoming Appt. Screen, Cancel Appt. Screen, Appt. Cancelled Screen, Past Appt. Screen
Book Appointment	FR6: The student must be able to submit an appointment request.	Home Dashboard, All Practitioners Screen, Female Practitioners Screen, Book Appt. Screen, Confirm Appt. Screen, Appt. Booked Screen
Log Mood & Check-in	FR11, FR13: Create mood entries and complete emotional check-ins.	Mood Check-in Screen, Emotional Check-in Screen
Journaling	FR12: Create, update, and delete their own journal entries.	Journal Screen (List & Entry Form)
Emergency SOS	Critical safety feature (Approved Phase 1 scope)	Emergency Screen
View Wellness Resources	Access stress-management and mindfulness content	Wellness Resources Screen
Profile Management	Manage personal profile and view streaks	Profile Screen

Performed Initial Function Checks for flutter

Function or check	Result	Action required
Application launches successfully	Passed. The app loads the Splash screen and transitions to Login without crashing.	None
Main screens display correctly	Passed. UI matches the Figma designs (colors, icons, and layout are correct).	Fix minor text overflow on the "Presenting Concerns" screen for smaller devices.
Navigation works	Passed. Bottom navigation bar and back buttons route correctly between screens	Ensure the "Back to Home" button on the Appt. Booked screen clears the navigation stack.
Forms validate input	Passed. Registration and Intake forms prevent empty submissions and validate email formats	Add stricter validation for the Student Number field (must be exactly 9 digits).
Mobile application connects to Supabase	Passed. The Supabase Flutter SDK is initialized and successfully pings the backend.	Test saving Journal entries and Mood logs to their respective tables.
Authentication works, where required	In Progress. Supabase Auth is configured and login returns a session token.	Implement the "Forgot Password" email reset functionality using Supabase Auth.
Access restrictions work as intended	passed	

Developing the ASP.NET Web Application

Review the Approved Web Application Requirements

Web Function	Related requirement or use case	Required Page
Manage Practitioners (Admin)	FR10: Admin must be able to verify practitioners	Practitioners List Page, Add Practitioner Modal, Update Practitioner Modal, Confirm Deletion Modal
Staff Authentication	Staff Login (Admin, Therapist, Peer Educator)	Admin Login Page
View Admin Dashboard	Admin oversight and reporting	Admin Dashboard (showing metrics, trends, and system health)
Manage Appointments (Admin)	Admin manages all bookings and resolves conflicts	Admin Appointments Management Page
Manage Students (Admin)	Admin oversees student accounts and system data	Admin Student Management Page, Full Student Info Report Page
View Therapist Dashboard	Therapist views their daily schedule and alerts	Practitioner Dashboard (showing Today's Appointments, Pending Requests, Follow-ups)
View Authorised Student Info (Therapist)	FR9: Practitioner views authorised student wellness records	Student Information Page / Practitioner Dashboard panels
Refer to Therapist (Peer Educator)	Escalate a student case to professional staff	Referral Screen (Form to refer a student to the Wellness Centre)

Perform Initial Function Checks

Function or check	Result	Action required
Web application launches successfully	Passed. The ASP.NET application builds and runs successfully on localhost. The landing page routes to the Staff Login screen.	None.
Main pages display correctly	Passed. The Admin Dashboard, Practitioner Dashboard, Appointments list, and Peer Educator views match the Figma designs (layout, colors, and tables).	none

Navigation works	Passed. The sidebar correctly routes between Dashboard, Appointments, Practitioners, Students, and Messages.	Ensure the "Log Out" link correctly clears the session and redirects to the Login page.
Forms validate user input	Passed. The "Add Practitioner" and "Update Appointment" modals correctly block empty submissions.	Add stricter validation to the "Refer to Therapist" form (e.g., require a brief context description).
Web application connects to Supabase	Passed. The Supabase C# client is configured and successfully connects to the shared project using the API keys.	Move the Supabase URL and Key to appsettings.json and ensure this file is in .gitignore
Required records can be retrieved	Passed. The web app successfully fetches and displays appointments, practitioner lists, and student data from the Users and Appointments tables.	none
Records can be created or updated where required	Passed. Successfully updated an appointment status from "Pending" to "Confirmed" and added a new practitioner to the database.	Test the "Delete Practitioner" function to ensure foreign key constraints are handled gracefully.
Authentication works, where required	In Progress. Staff login is connected to Supabase Auth and returns a valid session.	Implement role-based redirects (e.g., Admins go to Admin Dashboard, Therapists go to Practitioner Dashboard).
User access restrictions work as intended	In Progress. Basic Role-Based Access Control (RBAC) is active (e.g., Peer Educators cannot access Admin pages).	Configure Row Level Security (RLS) in Supabase so Therapists can only view student records for students they are assigned to.

Integrate the Mobile and Web Applications

Shared process	Action in Flutter	Action in ASP.NET	Shared information
Process 1: Appointment Booking & Management	Student: Selects a practitioner, date, and time, then submits an appointment request.	PractitionersAdmin: Views the incoming request on their dashboard, then confirms, reschedules, or cancels the appointment.	Student ID, Practitioner ID, Appointment Date/Time, Slot ID, Status (Pending, Confirmed, Cancelled).
Process 2: Practitioner Availability Sync	Student: Views real-time available dates and time slots for a specific practitioner.	Therapist: Updates their availability by adding or removing time slots on the web dashboard.	Practitioner ID, Slot Date, Start Time, End Time, Is_Available
Process 4: Mood & Wellness Tracking	Student: Logs daily mood and completes emotional check-ins (stress/coping levels).	Therapist: Views the student's mood history graph and emotional check-ins on their dashboard (if authorised).	Student ID, Mood Level, Stress Level, Coping Level, Logged_at timestamp.

Confirmation that both applications use the same backend

Item to check	Flutter	ASP.NET
Correct Supabase project	Connected to the CampusCare_GroupF Supabase URL and Anon Key.	Connected to the exact same CampusCare_GroupF Supabase URL and Anon Key.
Required database tables	Reads/Writes to: Users, Student_Profile, Appointments, Mood_Entry, Availability_Slot,	Reads/Writes to: Users, Practitioner, Appointments, Availability_Slot, Risk_Flag.
Appropriate authentication	Uses Supabase Auth for Student login. Generates a JWT (JSON Web Token) upon successful login.	Uses Supabase Auth for Staff login. Generates a JWT upon successful login, verifying the role (Admin, Therapist, Peer Educator).
Required database operations	**Create:** Bookings, Moods, Messages. **Read:** Practitioner lists, available slots, own appointments. **Update:** Cancel own appointment, update profile.	**Read:** All student appointments, student wellness data (if authorised), **Update:** Appointment status, Practitioner availability. **Create:** New Practitioners (Admin).

Approved user permissions	RLS Enforced: A student can only SELECT, INSERT, and UPDATE rows where student_id matches their Auth UID. They cannot see other students' data.	**RLS Enforced:** Therapists and Peer Educators can only read student data assigned to them. Only Admins have the RLS bypass (Service Role) to manage practitioners and system-wide data.

Check Authentication and Access Across the System

User role	Expected action	Result
Mobile user (Student)	Access their permitted functions and information	Passed. Student successfully logged in
Web user (Therapist/ Admin)	Access their permitted functions and information	Passed. Therapist logged in and successfully retrieved the appointment booked by the student in the Flutter app. Admin logged in and successfully viewed the User Management and Practitioner Verification pages.
Unauthorised user	Attempt to access a restricted URL or database table without the correct role or session token.	Passed. An unauthenticated request to the ASP.NET /Admin/Dashboard redirects to the Login page. Direct API calls to Supabase without an auth token are rejected by RLS (Row Level Security) policies.

Record and Correct Integration Problems

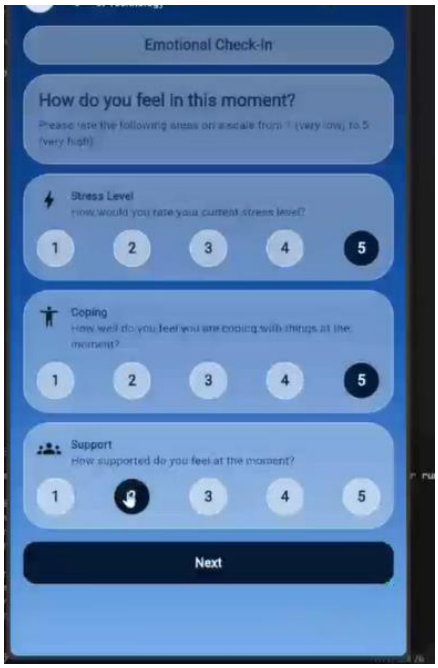
Integration problem	Possible cause	Action taken
New request does not appear in ASP.NET	Incorrect query or connection	Updated the ASP.NET C# LINQ query to fetch all appointments for the logged-in therapist regardless of status. Ensured the Flutter app sends the status as "Pending".
Updated status is not shown in Flutter	Information was not retrieved again	Added a "pull-to-refresh" function on the Flutter Upcoming Appointments screen, and ensured the initState() method re-queries Supabase every time the screen is opened.
A user cannot retrieve an authorised record	Incorrect access policy or query	Updated the Supabase RLS policy for the Messages table to allow access if the user_id matches the conversation_id participant. Fixed the ASP.NET query to join on Conversation_ID.

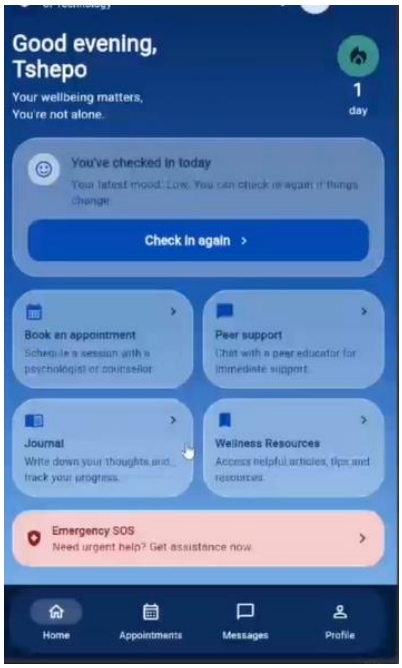
Recordings of integration

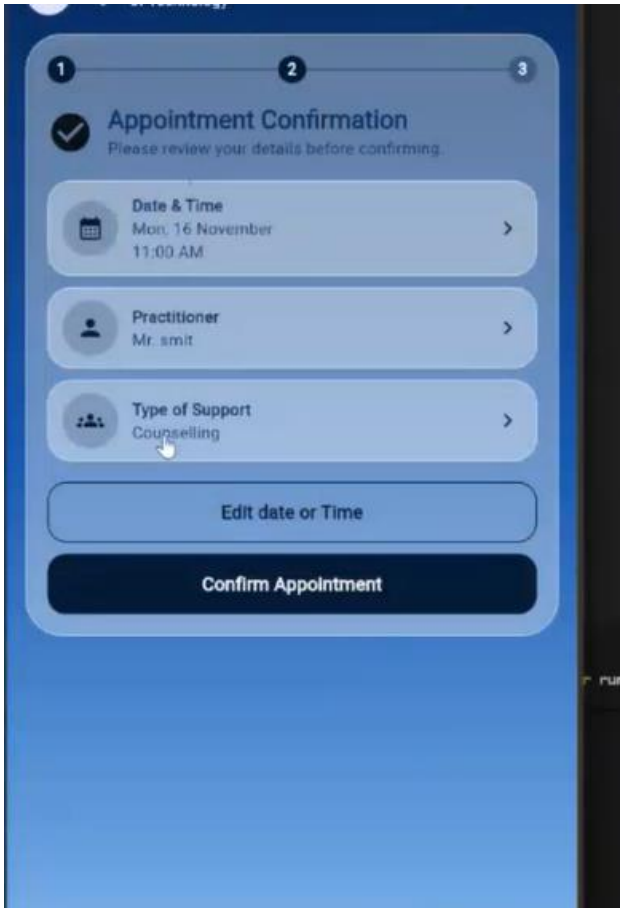
Demonstrate the Integrated Prototype Link to prototype video:https://drive.google.com/file/d/1QOgHV4cjtFLoF2YU_jLpl7BxVZCEiJqg/view?usp=sharing

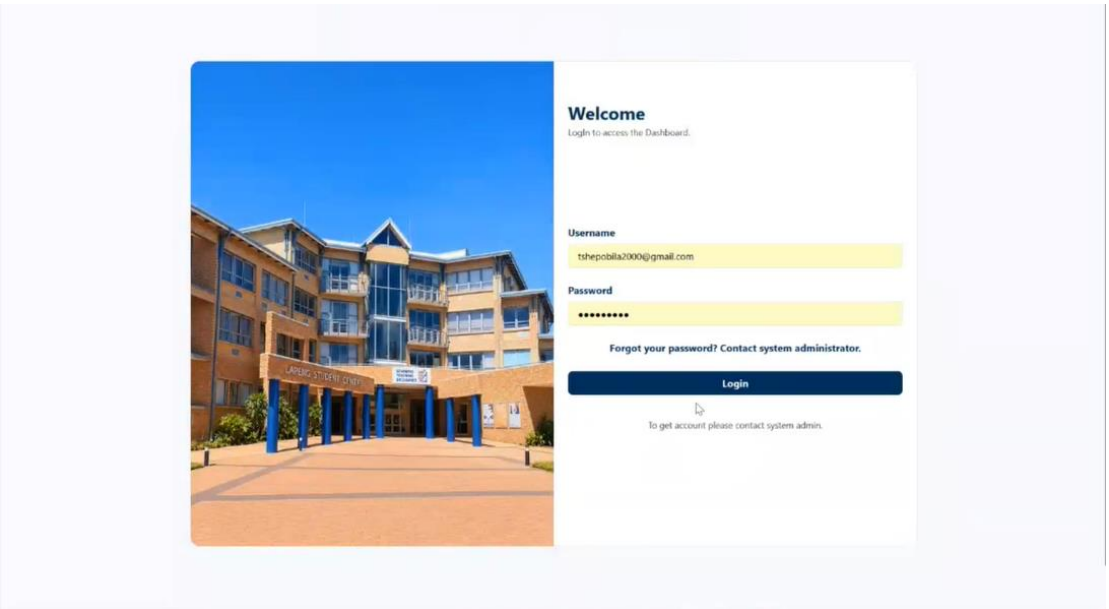
The Flutter mobile application running link :https://drive.google.com/file/d/1X6EYKVp0B8QNwQE1_p9aFmz4ehPuwbL1/view

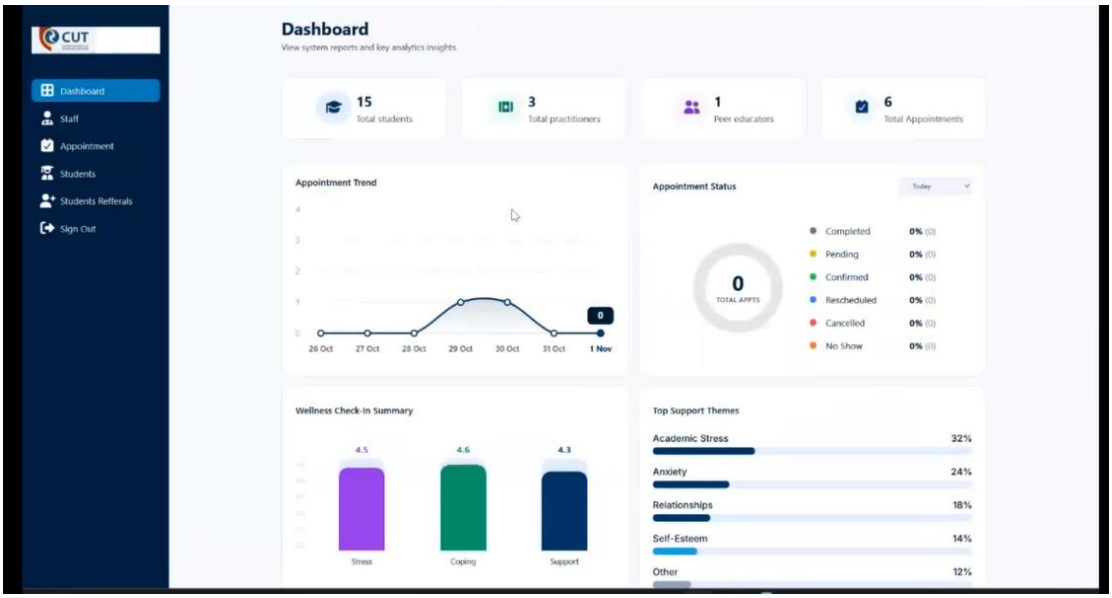
The ASP.NET web application running :https://drive.google.com/file/d/1kQqyNTX4qrNAv_-ogBHOWG7qZiLgqisO/view

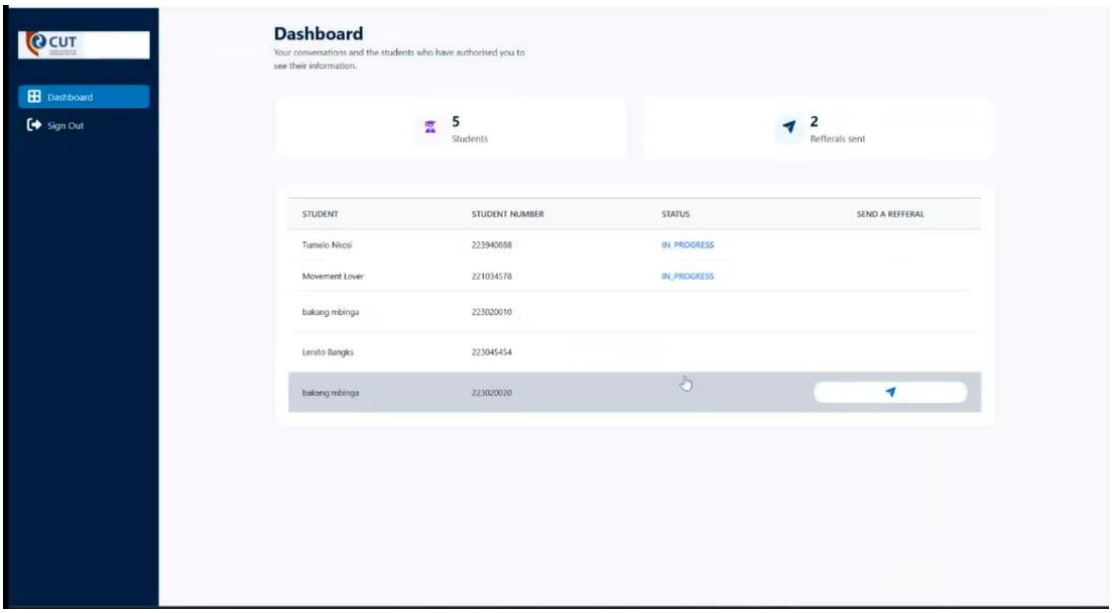




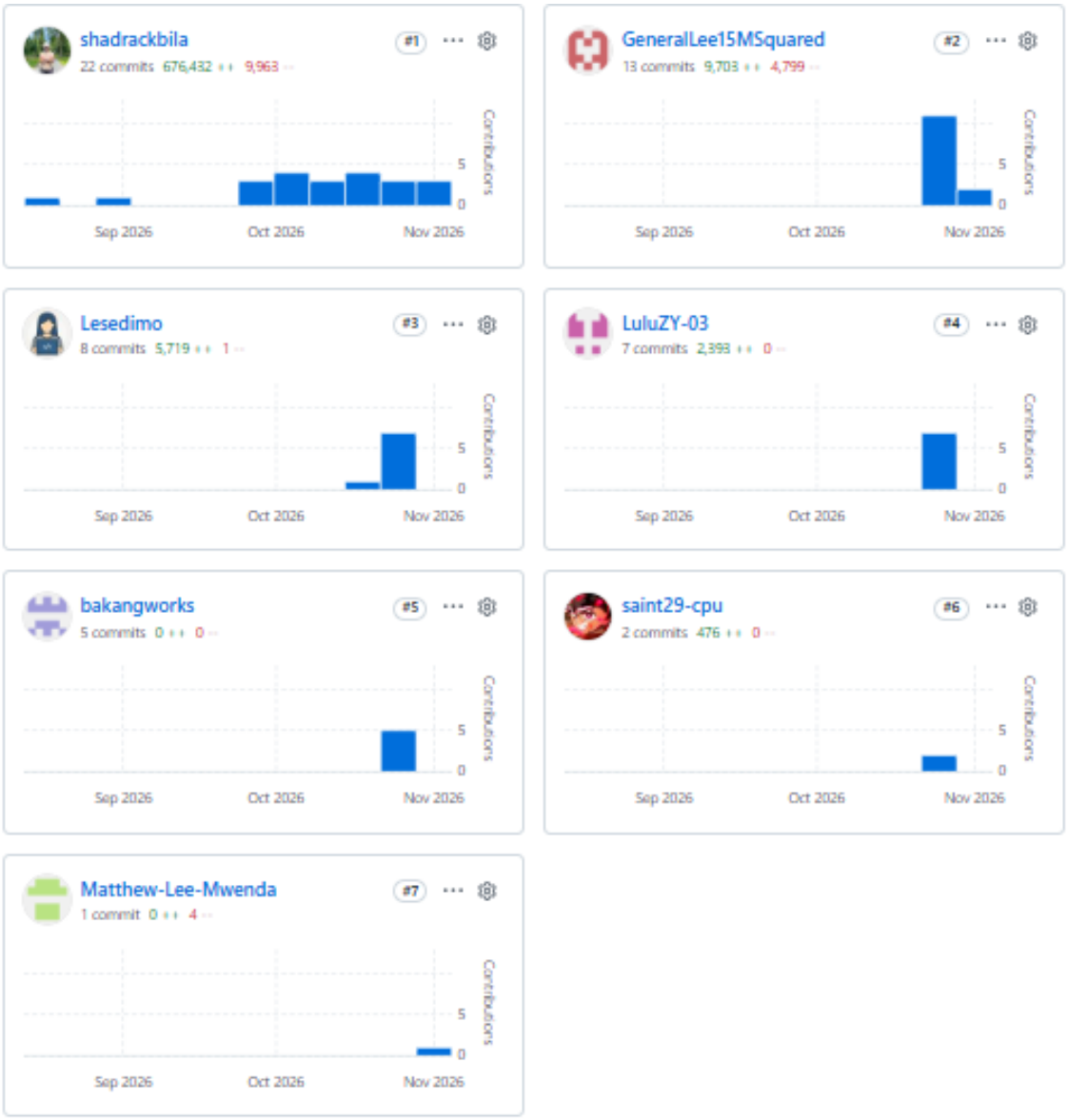



Welcome
Login to access the Dashboard.
Username
tshepobila2000@gmail.com
Password
Forgot your password? Contact system administrator.
Login
To get account please contact system admin.

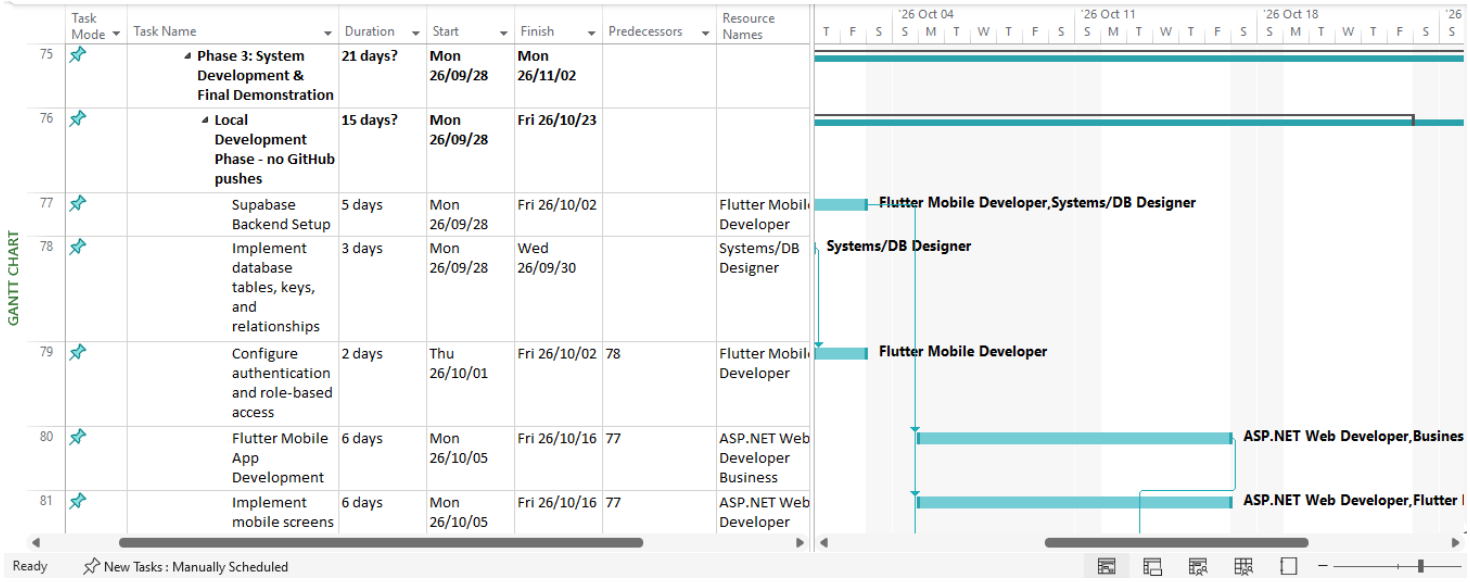

CUT
Dashboard
Staff
Appointment
Students
Students Refferals
Sign Out
Dashboard
View system reports and key analytics insights.
15
Total students
3
Total practitioners
1
Peer educators
6
Total Appointments
Appointment Trend
26 Oct
27 Oct
28 Oct
29 Oct
30 Oct
31 Oct
1 Nov
Appointment Status
0
TOTAL APPTS
Completed 0%
Pending 0%
Confirmed 0%
Rescheduled 0%
Cancelled 0%
No Show 0%
Wellness Check-In Summary
4.5
4.6
4.3
Stress
Coping
Support
Top Support Themes
Academic Stress 32%
Anxiety 24%
Relationships 18%
Self-Esteem 14%
Other 12%


CUT
Dashboard
Sign Out
Dashboard
Your conversations and the students who have authorised you to see their information.
5
Students
2
Refferals sent
STUDENT
STUDENT NUMBER
STATUS
SEND A REFFERAL
Tumelo Nkosi
223940888
IN_PROGRESS
Movement Lover
221034578
IN_PROGRESS
bakang mbinga
223020010
Lerato Bangks
223045454
bakang mbinga
223020020

Evidence of GitHub Commits



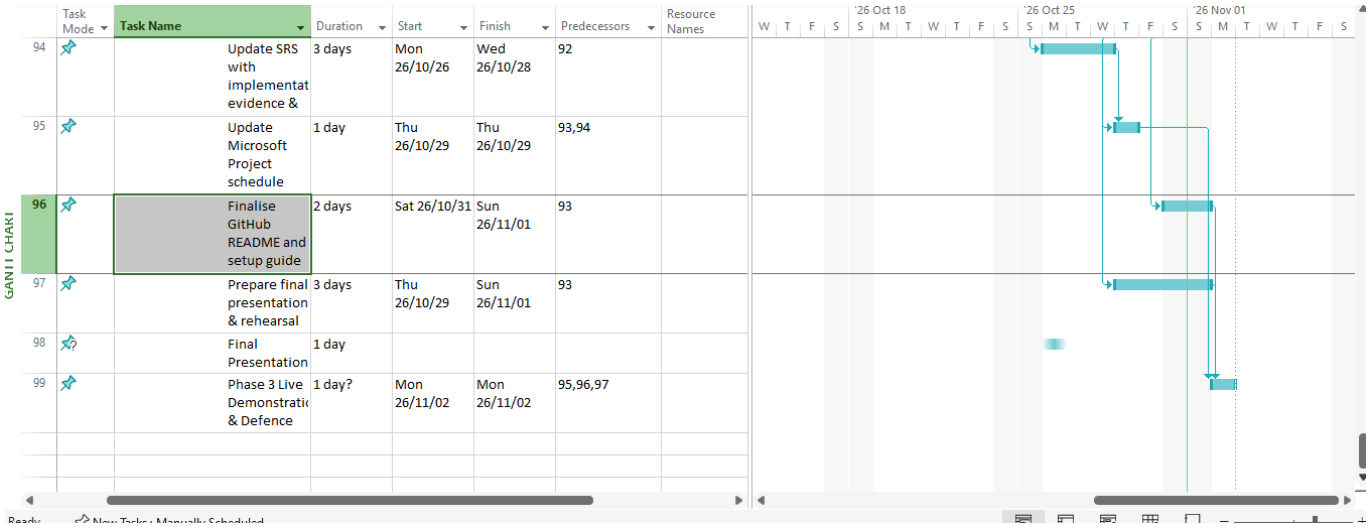
Final Updated Microsoft Project



	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
75		**Phase 3: System Development & Final Demonstration**	**21 days?**	**Mon 26/09/28**	**Mon 26/11/02**		
76		**Local Development Phase - no GitHub pushes**	**15 days?**	**Mon 26/09/28**	**Fri 26/10/23**		
77		Supabase Backend Setup	5 days	Mon 26/09/28	Fri 26/10/02		Flutter Mobil Developer
78		Implement database tables, keys, and relationships	3 days	Mon 26/09/28	Wed 26/09/30		Systems/DB Designer
79		Configure authentication and role-based access	2 days	Thu 26/10/01	Fri 26/10/02	78	Flutter Mobil Developer
80		Flutter Mobile App Development	6 days	Mon 26/10/05	Fri 26/10/16	77	ASP.NET Web Developer Business
81		Implement mobile screens	6 days	Mon 26/10/05	Fri 26/10/16	77	ASP.NET Web Developer

	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
82		ASP.NET Web App Development	6 days	Mon 26/10/05	Fri 26/10/16	77	ASP.NET Web Developer Business
83		Implement web pages and navigation (Local)	6 days	Mon 26/10/05	Fri 26/10/16	77	Risk Analyst, UI/UX Lead
84		Stakeholder Consultation	1 day	Wed 26/10/14	Wed 26/10/14	80,82	ASP.NET Web Developer
85		Showcase local builds to Cleo Maluka & gather	1 day	Wed 26/10/14	Wed 26/10/14	80,82	ASP.NET Web Developer Business Analyst
86		Local Integration & Testing	5 days	Mon 26/10/19	Fri 26/10/23	84	Project Manager Systems/DB
87		Connect Flutter to Supabase (Local)	3 days	Mon 26/10/19	Wed 26/10/21	84	Flutter Mobil Developer
88		Connect ASP.NET to Supabase (Local)	3 days	Mon 26/10/19	Wed 26/10/21	84	Risk Analyst

89		Test Shared Process 1 (Appointment Booking)	2 days	Thu 26/10/22	Fri 26/10/23	87,88	Project Manager UI/UX Lead
90		Test Shared Process 2 (Status Update)	2 days	Thu 26/10/22	Fri 26/10/23	87,88	Project Manager Risk Analyst
91		**Version Control & Finalisation Phase**	**6 days?**	**Mon 26/10/26**	**Mon 26/11/02**		
92		Data consistency check across apps	1 day	Mon 26/10/26	Mon 26/10/26	89,90	Systems/DB Designer
93		Push all local code to GitHub repository	6 days	Mon 26/10/26	Sat 26/10/31	92	ASP.NET Web Developer Business Analyst
94		Update SRS with implementat evidence &	3 days	Mon 26/10/26	Wed 26/10/28	92	



Final Assessment check point

Check the Integration of the Applications

Shared process	Flutter action	ASP.NET action	Working?
Your process 1: Appointment Status Update	Student opens the "Upcoming Appointments" tab to view the status of their booked sessions.	Therapist logs in, reviews a pending appointment, and changes the status from "Pending" to "Confirmed" (or "Rescheduled").	Yes. The therapist's action updates the Appointments table in Supabase. When the student refreshes their Flutter app, the app retrieves the new data and displays the "Confirmed" status.
Your process 2: Appointment Booking	Student selects an available practitioner and time slot, then confirms the booking request.	Therapist logs into the dashboard, sees the new "Pending" appointment in their queue, and confirms the booking.	Yes. The student's booking instantly appears on the therapist's dashboard. The therapist's confirmation updates the database, which the student sees on their next app refresh.

Checked Alignment With the Approved Requirements

Approved requirement	Application	Implemented function	Working?
FR1: Student register and log in	Flutter	Registration & Login screens connected to Supabase Auth.	Yes
FR2 & FR5: Complete intake form & provide consent	Flutter	Multi-step onboarding (Consent, Personal Info, Next of Kin, Presenting Concerns) saving to database.	Yes
FR6: Submit appointment request	Flutter	Book Appt. Screen with practitioner selection and available time slots.	Yes
FR7: View and manage appointments	Flutter	Upcoming/Past Appts. Screen allowing students to view and cancel bookings.	Yes
FR11 & FR13: Log mood & emotional check-in	Flutter	Mood Check-in and Emotional Check-in screens saving data.	Yes
FR12: Create journal entries	Flutter	Journal Screen with text input and mood tagging.	Yes
Emergency SOS Feature	Flutter	Emergency Screen with direct call/chat options.	Yes
FR8: Practitioner manage availability	ASP.NET	Therapist dashboard to add/remove available time slots.	Partial (Saving slots to DB works, but front-end calendar UI still needs polishing).
FR9: Practitioner view authorised student info	ASP.NET	Therapist dashboard to view student mood history and check-ins.	Yes
FR10: Admin verify practitioners	ASP.NET	Admin "Manage Practitioners" page with verification status toggle.	Yes

Appointment Reminders (FR14)	ASP.NET	Basic push notification setup (Simplified from Phase 2).	No (Outstanding: The automated email/push notification service is postponed due to time constraints).

Check Outstanding Development Problems

Problem or outstanding task	Action required	Responsible Member	Status
Automated Appointment Reminders (FR14): The push notification/email reminder system was not fully implemented.	Record as a known limitation in the SRS. The current prototype relies on the student manually checking the app. The automated cron-job can be built in Phase 4/future work.	(Feature Lead)	Outstanding (Recorded as Limitation)
secure Messaging (Requirement Change): The secure messaging feature between Students and Peer Educators was removed from the final prototype.	Record as an approved scope change in the SRS. The database tables (Conversations, Messages) remain in Supabase but the frontend UI was removed to prioritize core booking and mood tracking functions within the timeframe.	(Feature Lead)	Outstanding (Recorded as Limitation/Change)

GitHub link : https://github.com/shadrackbila/CampusCare_WIL_Group-F.git